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MASTER THESIS

CityHealth: Design and Development of an Intelligent Urban Healthcare

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Dedication

First and foremost, we dedicate this thesis to our beloved parents, whose unconditional love, constant guidance, and countless sacrifices have shaped our academic journey and strengthened our determination. We also extend a heartfelt thank you to our brothers for their continuous encouragement and support.

We share this dedication with our fellow researchers and classmates who walked this rigorous path alongside us. May our collective pursuit of learning continue to inspire innovation and service to our community.

Finally, we dedicate this milestone to ourselves. May this work remain a lasting reminder of what dedication, patience, partnership, and collaborative effort can accomplish.

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Naimi Abdeldjalil
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Abstract

Problem: The city of Sidi Bel Abbès, home to over 600,000 inhabitants, currently lacks a centralized digital healthcare infrastructure. Despite Algeria’s high internet penetration rate of 76.9% with over 54 million active mobile subscriptions, patients struggle with a severe information deficit. Citizens rely on fragmented, unverified sources to find medical help, resulting in prolonged search times, unnecessary physical travel, and dangerous delays during medical emergencies.

Solution: To resolve this issue, we developed CityHealth, a centralized and verified digital health ecosystem. This web platform aggregates local clinics, doctors, pharmacies, and laboratories into a single, accessible interface. The system integrates an intelligent multi-criteria search engine, an interactive map for real-time geolocation of emergency services, and a 24/7 multilingual Artificial Intelligence assistant (Arabic, French, English) to guide and triage patients.

Added Value: As an innovative startup initiative, CityHealth’s primary impact is reducing the time required to find a healthcare provider by up to 80%. It bridges the digital divide by offering a 100% free, reliable tool for citizens while providing a verified digital presence for over 500 local healthcare professionals. Ultimately, CityHealth transforms a historically fragmented system into a structured, inclusive digital health infrastructure, significantly improving healthcare accessibility in the region.

Keywords: E-Health, Artificial Intelligence, Healthcare Infrastructure, Web Platform, Sidi Bel Abbès.

Contents

Dedication	i
Acknowledgements	ii
Abstract	iii
Table of Contents	vii
List of Figures	x
List of Tables	x
List of Acronyms and Abbreviations	xi
General Introduction	1
I Theoretical Background and Scientific Implementation	3
1 Theoretical Context and Fundamental Concepts	4
1.1 Introduction	4
1.2 Web Development	4
1.2.1 Definition of Web Development	4
1.2.2 Modern vs Traditional Web Development	4
1.2.3 Front-end, Back-end and Full-Stack Development	5
1.3 Backend Development	5
1.3.1 Server-Side Programming	5
1.3.2 Frameworks	6
1.4 Artificial Intelligence	7
1.4.1 What is Artificial Intelligence?	7
1.4.2 How Does Artificial Intelligence Work?	7
1.5 System Communication and Integration	9
1.5.1 Application Programming Interface (API)	9
1.5.2 Model Context Protocol (MCP)	10
1.6 Technologies and Tools Used	11
1.6.1 Node.js	11
1.6.2 Express.js	11

1.6.3	React	11
1.6.4	Vite	11
1.6.5	Supabase	11
1.6.6	Tailwind CSS	12
1.7	Chapter Conclusion	12
II	State of the Art and Related Work	13
2	State of the Art and Related Works	14
2.1	Chapter Introduction	14
2.2	Scientific Literature Review	14
2.3	Study of Existing Solutions (Market Analysis)	15
2.3.1	Tabibi DZ	15
2.3.2	Sihhatech	15
2.3.3	Dwaya	15
2.3.4	Social Media Communities (Facebook)	15
2.3.5	Google Maps	15
2.4	Limitations of the Existing Solutions	16
2.5	Positioning of CityHealth (Competitive Advantage)	16
2.6	Conclusion of the Chapter	17
III	Modelling and Methodological Approach	18
3	Preliminary Study and System Design	19
3.1	Chapter Introduction	19
3.2	Merise Methodology	19
3.2.1	Conceptual Data Model (MCD)	19
3.2.2	Logical Data Model (MLD)	20
3.2.3	Physical Data Model (MPD)	21
3.3	Unified Modeling Language (UML)	22
3.3.1	Use Case Diagram	22
3.3.2	Activity Diagram	23
3.3.3	General Class Diagram	25
3.3.4	Sequence Diagrams	26
3.4	Technical Architecture and Security Model	29
3.4.1	Introduction	29
3.4.2	Global Technical Architecture	29
3.4.3	Defense-in-Depth Security Framework	29
3.5	Chapter Conclusion	31

IV	Scientific Implementation and Results	32
4	Scientific Implementation and Results	33
4.1	Introduction	33
4.2	Work Environment and Tools	33
4.3	Prototype (MVP) Realisation	34
4.3.1	Core Platform & Landing Interfaces	34
4.3.2	Authentication & User Onboarding	35
4.3.3	User Dashboards & Provider Ecosystem	37
4.3.4	Geospatial & Search Capabilities	39
4.3.5	Emergency Response & Community Mutual Aid	41
4.3.6	AI Integration & Medical Research	43
4.3.7	Developer API & Support Infrastructure	44
4.3.8	Mobile Application	46
4.4	Tests, Experiments and Results	58
4.4.1	Test Protocol	58
4.4.2	Results Obtained	59
4.5	Discussion of Results	59
4.6	Conclusion of Part I	59
V	Economic Dimension — Startup Project	60
5	Startup Business Model and Financial Study	61
5.1	Introduction to Part II	61
5.2	Business Model Canvas	61
5.3	Technical Sheet and Financial Study	62
5.3.1	Investments (CAPEX)	63
5.3.2	Revenue Projections	63
5.3.3	Direct Purchases — Digital Infrastructure	64
5.3.4	Payroll	64
5.3.5	External Expenses	65
5.3.6	Financing Summary	65
5.4	Conclusion of Part II	66
VI	General Conclusion and Perspectives	67
	General Conclusion and Perspectives	68

VII	Bibliographie	71
	Bibliography	72

List of Figures

1.1	General Architecture of an API	9
1.2	General Architecture of the Model Context Protocol (MCP)	10
3.1	Conceptual Data Model (MCD) of the CityHealth Platform	20
3.2	Logical Data Model (MLD) of the CityHealth Platform	20
3.3	Physical Data Model (MPD) of the CityHealth Database	21
3.4	Global Use Case Diagram for the CityHealth Platform	23
3.5	Activity Diagram — Medical Publication Submission Pipeline	24
3.6	General Class Diagram of CityHealth	25
3.7	Sequence Diagram — User Authentication Flow	26
3.8	Sequence Diagram — Appointment Lifecycle	27
3.9	Sequence Diagram — Provider OCR Verification Flow	28
4.1	The CityHealth main landing page, featuring an intuitive, centralized search engine for medical professionals in Sidi Bel Abbès, complemented by quick-access specialty tags and real-time platform statistics.	34
4.2	A visual guide outlining the seamless user journey on the platform, illustrating the three-step process for patients: searching verified databases, comparing profiles, and booking appointments.	34
4.3	An interactive interface showcasing the comprehensive suite of CityHealth services, extending beyond medical booking to include critical community features such as blood donation alerts and medical equipment sharing.	35
4.4	The comprehensive platform footer, highlighting complete multilingual support (i18n), seamless mobile application integration, and dynamic real-time platform statistics.	35
4.5	The secure citizen authentication portal, demonstrating the integration of modern access methods including OAuth providers and passwordless Magic Links via Supabase Auth.	36

4.6	The dedicated authentication gateway for healthcare professionals, enforcing strict Role-Based Access Control (RBAC) and tailored specifically for medical providers.	36
4.7	The dedicated onboarding portal for healthcare professionals, detailing the streamlined registration, profile completion, and strict verification process required to join the medical directory.	37
4.8	The centralized patient dashboard, featuring a dynamic profile completion tracker, secure appointment management, and quick access to emergency and donation records.	37
4.9	Provider onboarding interface	38
4.10	The public-facing profile of a verified healthcare provider, displaying critical operational details, facility information, and an aggregated patient review score.	38
4.11	The lower section of the provider profile, highlighting the transparent patient review system, available consultation slots, and a localized mini-map for precise navigation.	39
4.12	The tiered subscription model interface for healthcare providers, clearly delineating the functionalities of the Basic, Standard, and Premium plans, including advanced dashboard analytics and AI-driven visibility.	39
4.13	The interactive geolocation map interface, utilizing Leaflet to display verified healthcare providers in Sidi Bel Abbès, complete with dynamic category filtering and real-time operational status.	40
4.14	The geospatial search interface, allowing patients to dynamically locate nearby clinics and healthcare professionals using interactive map markers and advanced filtering criteria.	40
4.15	Advanced geolocation and routing integration within the interactive map, providing patients with real-time distance calculations and optimal itineraries to healthcare facilities.	41
4.16	The dedicated blood donation and emergency response dashboard, designed to broadcast urgent blood requirements in real-time and display critical community engagement statistics.	41
4.17	The personalized Medical Emergency Card, generating a unique QR code that grants first responders instant access to critical patient data, with built-in functionality for local PDF exportation.	42
4.18	The community mutual aid and donation board, allowing verified users to locally offer or request medical supplies, medications, and health-related assistance.	42

4.19	The Professional Announcements board, providing a moderated communication channel for verified healthcare providers to publish medical news, clinic updates, and sponsored health services.	43
4.20	Interactive community forum	43
4.21	The AI-powered medical assistant interface, utilizing large language models (LLMs) to perform initial symptom triage and intelligently orient patients to the most appropriate healthcare specialists.	44
4.22	The dedicated Medical Research Hub, functioning as an academic repository where professionals can publish, search, and access verified scientific articles and clinical findings.	44
4.23	The CityHealth Public API developer portal, illustrating secure integration endpoints, SHA-256 API key authentication, and expected JSON payload structures for third-party health applications.	45
4.24	The tiered API quota management and pricing interface, detailing the specific rate limits (requests per day) and monetization strategy implemented for third-party developer access.	45
4.25	The comprehensive CityHealth documentation hub, featuring structured navigation for different user roles and an integrated AI assistant widget to provide contextual, real-time guidance.	46
4.26	The centralized user support and contact portal, integrating a secure, serverless messaging form and a dynamic Frequently Asked Questions (FAQ) module to streamline platform assistance.	46
4.27	Primary mobile home screen introducing the CityHealth ecosystem, designed to make access to core services and navigation between features immediately understandable for new users.	47
4.28	Secondary home view highlighting the main care pathways and shortcuts, simplifying entry to blood donation, emergency care, and service discovery directly from the landing screen.	48
4.29	Unified search interface for providers and clinics, enabling citizens to quickly look up medical professionals, facilities, and health services using a familiar search experience.	49
4.30	second unified search interface for providers and clinics, enabling citizens to quickly look up medical professionals, facilities, and health services using a familiar search experience.	50
4.31	Advanced filtering panel, allowing refinement of search results by availability, rating, spoken languages, and accessibility options to better match patient constraints.	51

4.32	Interactive map view displaying healthcare providers, emergency services, and blood donation centers around Sidi Bel Abbès, with real-time category filters for rapid orientation.	52
4.33	Embedded AI medical assistant, offering a conversational interface for initial symptom triage and intelligent redirection towards the most relevant health services.	53
4.34	Settings interface centralizing language selection, dark mode, access to documentation, FAQs, and the developer space, ensuring full control over the mobile experience.	54
4.35	Dedicated blood donation module summarizing key statistics and urgent calls, encouraging citizens to participate in life-saving donation campaigns directly from their phone.	55
4.36	Mobile-to-web bridge, presenting services that are currently only available on the CityHealth web platform and redirecting users seamlessly via QR code.	56
4.37	Notification activation flow, enabling users to opt in to real-time alerts for appointments, messages from healthcare providers, and critical platform notifications.	57
4.38	Professional announcements feed reserved for subscribed and verified providers, used to publish sponsored health offers, clinic updates, and public health campaigns to the mobile audience.	58
5.1	Business Model Canvas of CityHealth	62

List of Tables

2.1	Comparative Analysis of E-Health Solutions in Algeria	17
4.1	Hardware specifications of the collaborative development environment	33
4.2	Summary of prototype performance	59
5.1	Investment Plan — Initial Capital Requirements	63
5.2	Projected Revenue Streams — Year 1	64
5.3	Direct Purchase Costs — Digital Infrastructure (Year 1)	64
5.4	Payroll Structure — Year 1	65
5.5	External Operating Expenses (Year 1)	65
5.6	Financing Synthesis — Profitability Evaluation (Year 1)	66

List of Acronyms and Abbreviations

Acronym	Definition
API	Application Programming Interface
BMC	Business Model Canvas
MVP	Minimum Viable Product
RSSI	Réseaux, Systèmes et Sécurité de l'Information

General Introduction

General Context

Over the last decades, technology has significantly transformed the way people access information and services. Many sectors have benefited from digitalization, including education, commerce, transportation, and communication. However, access to healthcare information remains challenging in many regions.

The healthcare sector plays a fundamental role in society, as every individual requires medical services at different stages of life. With the continuous growth of healthcare facilities, medical specialties, and healthcare providers, finding the right service has become increasingly complex. This challenge is particularly noticeable in the city of Sidi Bel Abbès, where healthcare information is often scattered across different sources.

To address this issue, this thesis focuses on the design and development of **CityHealth**, an intelligent digital healthcare platform aimed at simplifying access to healthcare services and improving the visibility of local healthcare providers.

Problem Statement

Despite the advancement of healthcare services and medical specialization, many citizens still face difficulties when searching for appropriate healthcare providers. In most cases, people rely on traditional methods such as asking relatives, friends, or searching randomly for clinics and doctors. These approaches are often time-consuming, inefficient, and unreliable, especially during emergency situations.

Furthermore, there is currently no centralized digital healthcare platform dedicated to the Wilaya of Sidi Bel Abbès that provides comprehensive information about doctors, clinics, pharmacies, laboratories, and other healthcare services. This lack of digital infrastructure creates a gap between citizens and healthcare providers, resulting in wasted time, unnecessary travel, and limited access to reliable medical information.

Objectives of the Study

The main objective of this work is to design and implement **CityHealth**, an intelligent healthcare ecosystem that facilitates access to healthcare services through modern digital technologies.

From a scientific perspective, the project aims to develop a web-based platform that integrates:

- An interactive map for locating healthcare providers and facilities.
- An intelligent multilingual assistant available 24 hours a day.
- Advanced search and filtering functionalities.
- A centralized healthcare information system.
- A user-friendly interface accessible to all citizens.

From an entrepreneurial perspective, the project aims to establish a startup capable of contributing to the digital transformation of healthcare services in Algeria.

The proposed solution seeks to provide several benefits:

- Immediate access to local healthcare services for patients and citizens.
- Reliable guidance in emergency situations.
- Increased online visibility for healthcare providers.
- Better management of appointments and patient interactions.
- Reduction of healthcare search time and unnecessary travel.
- Contribution to the digitalization of healthcare services in the Wilaya of Sidi Bel Abbès.

Thesis Structure

This thesis is structured into two main parts.

Part I presents the theoretical background, related works, methodology, system design, implementation process, and experimental results of the proposed solution.

Part II focuses on the entrepreneurial aspect of the project. It presents the Business Model Canvas, the startup strategy, and the financial feasibility study that evaluates the economic viability of the proposed startup.

Finally, the thesis concludes with a summary of the achieved results, the main contributions of the project, and future perspectives for the improvement and expansion of the CityHealth platform.

PART I

Theoretical Background and Scientific Implementation

CHAPTER 1

Theoretical Context and Fundamental Concepts

1.1 Introduction

This chapter is about the definition of most of the technologies used in the development of this project.

1.2 Web Development

1.2.1 Definition of Web Development

Web development is the process of building, creating, and maintaining websites behind the scenes to make it run smoothly and more importantly “correctly” with a seamless user experience. Web developers “or WebDevs for short” use a variety of different programming languages and tools in this highly demanded field which includes aspects such as web design, web publishing, web programming, and database management, and it is generally broken down into front-end (the user-facing side) and back-end (the server side).

1.2.2 Modern vs Traditional Web Development

Web development and mobile app development has changed considerably in past few years due to the huge advancement in technology. Few years Ago, client side machines were considered very much limited in power and resources to handle huge amount of information calculation, so the solution was to equip a high-end server to handle all of the logic and the work load of the application on the server-side, and only transfer HTML pages to the client-side with some minor JavaScript form and input validation scripts.

With the advancement of technology, the information processing capabilities of the client-side machines grew significantly. Alongside this advancement, new web technologies and frameworks were released on the market replacing the old fashioned features that would be found only on the server side such as JSP, ASP with new, more pow-

erful JavaScript client-side frameworks like Angular, Backbone, Underscore etc. This influenced the shift of processing from the server side more to the client side.

In Modern web development architecture, Restful API's on the server of the application sends data to the client upon request, the client side frameworks which provide features like 2-way data binding, master page layout, routing and descriptive HTML ect, create dynamic pages on the fly with the requested data. Modern web development architecture allows us to process more requests per second than ever due to the smaller load on the server. We can scale our application very easily.

1.2.3 Front-end, Back-end and Full-Stack Development

Front-end and back-end development are the key components that keep the system running at peak efficiency in applications and websites. Front-end operations deal with the system interface while the back-end is usually behind the scenes dealing with servers, databases and other applications that provide information to the user.

Since the back-end and the front-end are different aspects of development, they have become significantly specialized. Each has very specific programming languages and frameworks that require focused technical expertise. So this is the etiology of highly specialized front-end and back-end developers who focus on specific parts of front-end and back-end development, respectively. There is yet another unique set of developers; full-stack developers. These are well placed to handle every aspect of back-end and front-end development.

1.3 Backend Development

1.3.1 Server-Side Programming

Back-end architects design and build the systems, data layers, and other services that support a website. As such, back-end developers can also work with various algorithms in a website as well as front end developers to create solutions that provide better user interaction and satisfaction.

Back-end programming is indeed the cornerstone of major content management systems such as WordPress as well as most complex web applications in use today. In fact, it's not wrong to think that it's through back-end programming that you find solutions and build whatever you want. Data sets and algorithms are the backbone of back-end coding. Database technologies which are SQL variants are therefore quite indispensable when building back-end models and database queries.

Back-end developers use different languages in their work. While JavaScript is still very popular in back-end development, it has recently fallen out of favor with most

back-end programmers. Popular languages include Python, Ruby / Rails, and PHP, among others.

1.3.2 Frameworks

A framework is a set of tools and software components organized in accordance with an architecture plan and patterns. It is often provided in the form of a software library, and accompanied by the plan of the target architecture of the framework. A framework is designed to help programmers in their work. The organization of the framework aims at the maximum productivity of the programmer who will use it - reduction of the costs of construction and maintenance of the program. The exact content of the framework is dictated by the type of program and the target architecture for which it is designed.

Advantages of Using Frameworks

Using a framework ensures that the built application is better structured and more secure, thanks to components tested, approved and used by other developers. These components form software bricks that allow the developer not to reinvent the wheel. One of the big advantages of a framework is that it can be used for several projects. In addition, it contains ready-to-use components, which eliminates the need to build a website from scratch.

The developer can therefore focus only on the specific needs of the customer. All other common and obvious features such as securing forms, processing images, etc... are dispensed with them. Also, the use of a framework allows the establishment of good coding practices by requiring developers to respect a particular convention.

These conventions are in fact rules to be observed. For example, this could be a naming convention for variables, functions, or even classes. These rules allow developers to produce clean and understandable code. Nevertheless, a framework requires a certain mastery of the programming language. More precisely, having notions of OOP (Object Oriented Programming) is necessary to be able to develop properly with a framework. The frameworks are intended for experienced developers. Another advantage that frameworks allow is the possibility of working with several people on the same project. This facilitates collaboration and the realization of a group project.

1.4 Artificial Intelligence

1.4.1 What is Artificial Intelligence?

Artificial Intelligence (AI) refers to a collection of technologies that enable computers to learn, reason, and perform advanced tasks that previously required human intelligence. These tasks include understanding language, analyzing data, recognizing patterns, and providing useful recommendations. As a transformative technology, AI is driving significant changes across industries and positively impacting individuals, organizations, and society as a whole.

AI encompasses multiple disciplines, including computer science, data analysis, statistics, software and hardware engineering, linguistics, neuroscience, philosophy, and psychology. The interdisciplinary nature of AI allows researchers and developers to design systems capable of solving complex real-world problems.

The primary goal of AI is to enable computers to perform tasks that resemble human cognitive abilities, such as understanding their environment, learning from experience, and generating new insights. For example, AI is widely used in Optical Character Recognition (OCR) systems to extract text and information from documents and images. This process converts unstructured content into structured data that can be analyzed and utilized by organizations to support decision-making and generate valuable insights.

1.4.2 How Does Artificial Intelligence Work?

Although Artificial Intelligence includes a wide variety of techniques, all AI systems fundamentally rely on three main components: data, algorithms, and computational power. AI models learn and improve by processing large amounts of data, identifying patterns and relationships that may not be easily detectable by humans. Consequently, both the quality and quantity of training data play a crucial role in determining the performance of AI systems.

AI is not a single technology but rather a broad field that includes several important subdomains, as described below.

Machine Learning (ML)

Machine Learning is a branch of AI that enables systems to learn from data and identify patterns without being explicitly programmed for every task. By analyzing historical data, machine learning algorithms can make predictions or support decision-making processes. For instance, if a computer is shown thousands of images of birds, it can learn to recognize the characteristics of birds and identify them in new images.

Deep Learning (DL)

Deep Learning is a specialized subset of Machine Learning that uses artificial neural networks composed of multiple layers. Inspired by the structure and functioning of the human brain, these networks are particularly effective for complex tasks such as image recognition, speech recognition, and natural language understanding. The term “deep” refers to the large number of layers used within these neural networks.

Natural Language Processing (NLP)

Natural Language Processing (NLP) focuses on enabling computers to understand, interpret, and generate human language. NLP technologies power many modern applications, including virtual assistants, machine translation systems, chatbots, sentiment analysis tools, and text summarization systems.

Computer Vision

Computer Vision is the field of AI that enables computers to interpret and analyze visual information from the real world, such as images and videos. This technology is used in numerous applications, including facial recognition, medical image analysis, autonomous vehicles, surveillance systems, and quality control in manufacturing.

1.5 System Communication and Integration

1.5.1 Application Programming Interface (API)

An Application Programming Interface (API) is a set of rules and protocols that enables different software applications to communicate and exchange data, functionalities, and services with one another. APIs simplify software development by allowing developers to integrate existing services and capabilities instead of building them from scratch. They provide a secure and standardized mechanism through which applications can interact while hiding internal implementation details.

APIs play a fundamental role in modern software systems and are widely used in web applications, mobile applications, cloud services, and distributed systems. In the context of the CityHealth platform, APIs are used to facilitate communication between the frontend application, backend services, databases, and external healthcare-related services.

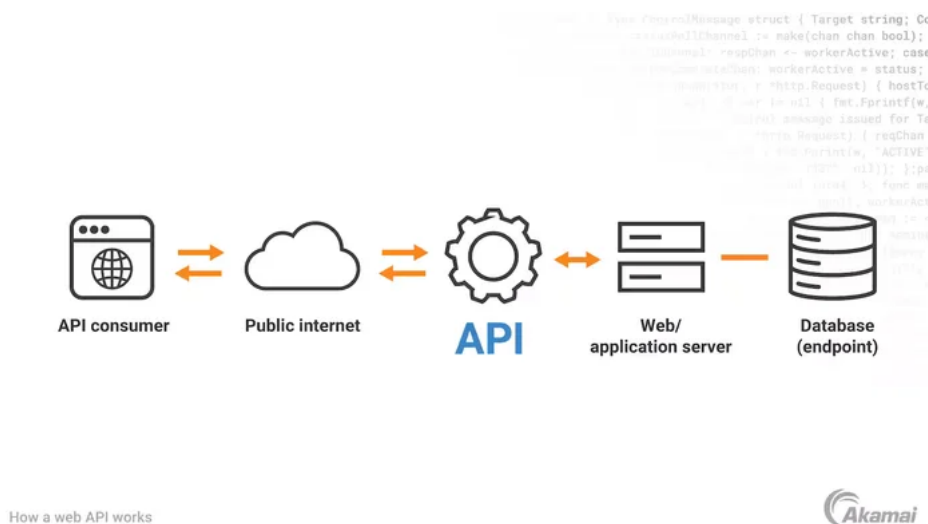


Figure 1.1: General Architecture of an API

1.5.2 Model Context Protocol (MCP)

The Model Context Protocol (MCP) is an open standard designed to enable seamless integration between Artificial Intelligence models and external data sources, tools, and applications. MCP provides a standardized interface through which AI systems can access information and perform actions without requiring custom integrations for each external service.

The protocol introduces a unified architecture that allows AI models to dynamically discover available tools, retrieve contextual information, and interact with external systems in a secure and structured manner. By standardizing communication between AI models and external resources, MCP improves interoperability, scalability, and maintainability in AI-powered applications.

In the CityHealth platform, MCP can facilitate communication between the AI assistant and various healthcare resources, enabling more intelligent and context-aware responses for users.

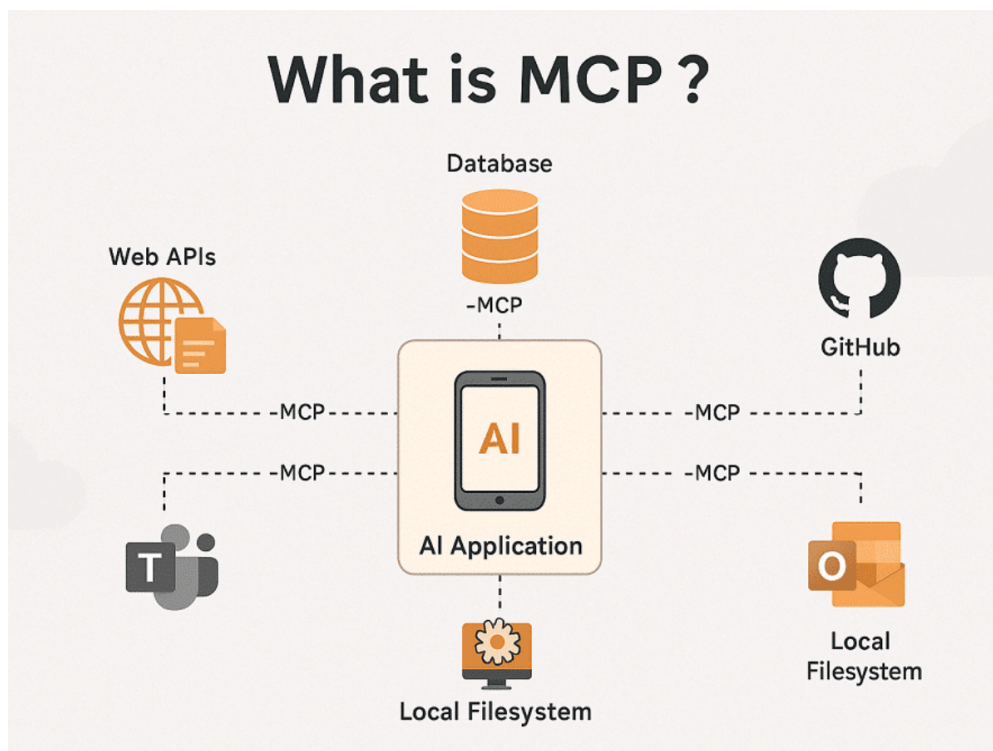


Figure 1.2: General Architecture of the Model Context Protocol (MCP)

1.6 Technologies and Tools Used

1.6.1 Node.js

Node.js is an open-source, cross-platform JavaScript runtime environment that executes JavaScript code outside a web browser. Built on Chrome's V8 JavaScript engine, it uses an event-driven, non-blocking I/O model that makes it lightweight and efficient, perfectly suited for data-intensive real-time applications that run across distributed devices.

1.6.2 Express.js

Express.js is a minimal, fast, and unopinionated web framework for Node.js. It provides a robust set of features to develop web and mobile applications, simplifying the process of writing server-side logic and building robust RESTful APIs. It is considered the de facto standard server framework in the Node.js *ecosystem*.

1.6.3 React

React is a declarative, efficient, and flexible open-source JavaScript library for building user interfaces or UI components. Maintained by Meta (formerly Facebook) and a community of developers, it allows developers to create large web applications that can change data, without reloading the page, by utilizing a virtual DOM for optimized rendering.

1.6.4 Vite

Vite (often referred to in the React ecosystem as React-Vite) is a modern frontend build tool that significantly improves the development experience. It consists of a fast development server that serves source files over native ES modules with lightning-fast Hot Module Replacement (HMR), and a build command that bundles code with Rollup for highly optimized production *assets*.

1.6.5 Supabase

Supabase is an open-source alternative to Firebase. It provides developers with a complete backend architecture out of the box, including a dedicated PostgreSQL database, auto-generated REST and GraphQL APIs, real-time data subscriptions, robust authentication, and cloud storage. It allows developers to build scalable applications quickly without the lock-in of proprietary NoSQL *ecosystems*.

1.6.6 Tailwind CSS

Tailwind CSS is a popular CSS framework that is entirely utility-first and provides users with low-level CSS classes in PostCSS that can be used to define components and designs independently. Tailwind CSS was developed by Adam Wathan and first released in 2017. Since then it's been installed by millions of users `users`.

1.7 Chapter Conclusion

This chapter presented the theoretical foundations and technological concepts required for the development of the CityHealth platform. It introduced the principles of web development, backend architectures, artificial intelligence technologies, system communication mechanisms, and the development tools used throughout the project. These concepts provide the necessary theoretical basis for the analysis, design, and implementation phases presented in the following chapters.

PART II

State of the Art and Related Work

CHAPTER 2

State of the Art and Related Works

2.1 Chapter Introduction

The transition toward digital healthcare is no longer a futuristic concept, but an urgent necessity. In order to design a robust, impactful, and innovative platform like CityHealth, it is imperative to deeply understand the current technological landscape. This chapter serves as the theoretical and analytical foundation of our project. First, we explore the recent scientific literature regarding the digital transformation of healthcare, with a specific focus on the Algerian context. Subsequently, we conduct a rigorous critical analysis of the existing digital solutions and applications currently operating within the national market. By dissecting the strengths and exposing the critical limitations of these platforms, we firmly establish the scientific positioning and the ultimate competitive advantage of CityHealth.

2.2 Scientific Literature Review

The integration of Information and Communication Technologies (ICT) in healthcare has been the subject of extensive recent research, particularly concerning developing nations striving to overcome infrastructural barriers.

Recent studies highlight a paradigm shift in the Algerian healthcare sector. According to Boufissiou (2024) in the analysis of the *Algerian Digital Health Strategy 2023-2027*, the COVID-19 pandemic accelerated the government's awareness of the necessity to modernize medical data management to reduce costs and improve service quality. However, research indicates that the current national health information system remains heavily decentralized and fragmented, creating a severe "Information Asymmetry" between healthcare providers and patients.

Furthermore, recent academic projects, such as the *Dual System for Hospital Emergency Placement* developed at the University Ziane Achour of Djelfa (2025), emphasize the critical need for real-time geolocation in emergency medical services. Their research demonstrates that providing immediate, accurate data regarding hospital capacities and available specialists drastically optimizes pre-hospital triage. Similarly, the integration of Artificial Intelligence (AI) in patient orientation is identified globally

as a vital tool to reduce waiting times and prevent hospital overcrowding. CityHealth aligns perfectly with these academic findings by proposing a solution that bridges the gap between theoretical digital strategies and localized, practical citizen implementation.

2.3 Study of Existing Solutions (Market Analysis)

To accurately assess the Algerian E-health market, we selected the most prominent digital solutions currently affecting the daily lives of Algerian citizens.

2.3.1 Tabibi DZ

Tabibi is currently one of the leading digital platforms for medical booking in Algeria. It provides a seamless interface allowing patients to search for doctors by specialty and Wilaya, offering real-time availability and instant booking confirmation via SMS and email. While highly efficient for planned consultations, its scope is strictly limited to doctor appointments.

2.3.2 Sihhatech

Sihhatech is a comprehensive Algerian health application offering remote medical appointments, clinic management dashboards, and a digital health record system. It focuses heavily on digitizing the internal management of clinics and pharmacies, offering robust tools for healthcare professionals to manage patient data.

2.3.3 Dwaya

Dwaya is an Algerian mobile application specifically designed to solve the pharmaceutical information gap. It excels in helping citizens locate nearby pharmacies, particularly duty pharmacies (pharmacies de garde), and provides information regarding medication availability.

2.3.4 Social Media Communities (Facebook)

In the absence of a unified official platform, Algerian citizens rely heavily on Facebook groups (e.g., "Sidi Bel Abbès Santé", "Pharmacie de garde SBA"). These community-driven networks are the primary fallback for emergency blood donation requests and finding open clinics during weekends.

2.3.5 Google Maps

While not an Algerian product, Google Maps is universally utilized by citizens to locate nearby hospitals, laboratories, and clinics via GPS.

2.4 Limitations of the Existing Solutions

Despite the undeniable utility of the aforementioned platforms, a critical analysis reveals major structural and functional gaps in the Algerian E-health ecosystem:

- **Extreme Fragmentation:** The current ecosystem forces citizens to juggle multiple applications. A patient must use *Tabibi* to book a doctor, *Dwaya* to find a pharmacy, and *Facebook* to request an emergency blood donation. There is no centralized "All-in-One" medical ecosystem.
- **Absence of Intelligent Triage (AI):** None of the active local platforms feature an Artificial Intelligence assistant capable of understanding a patient's symptoms (in local languages) and guiding them toward the correct medical specialty. Patients are left to guess which doctor they need.
- **Lack of Emergency Crisis Management:** Existing appointment apps completely ignore the "Emergency" dimension. During a critical health crisis at 3:00 AM, appointment booking systems are useless. The market lacks a platform with a dedicated, real-time map isolating 24/7 emergency centers and blood banks.
- **Data Reliability and Verification Risks:** Platforms like Facebook and Google Maps suffer from unverified data. A clinic listed as "Open" on Google Maps may be permanently closed, leading to dangerous delays during medical emergencies.
- **Lack of Hyper-Localization:** National apps often overlook Wilaya-level infrastructure. Sidi Bel Abbès requires a bespoke database that mirrors its real on-the-ground medical reality.

2.5 Positioning of CityHealth (Competitive Advantage)

CityHealth is engineered specifically to dismantle these limitations. It is not just another booking app; it is a **Unified Digital Health Ecosystem** designed exclusively for the Wilaya of Sidi Bel Abbès.

Our scientific and entrepreneurial positioning stands out through the following strategic advantages:

1. **The Centralization Paradigm:** CityHealth aggregates doctors, clinics, pharmacies, laboratories, and emergency services into a single, cohesive web platform.
2. **Multilingual AI Assistant:** We introduce a technological vanguard to the local market: an integrated AI assistant (answering in Arabic, French, and English)

that acts as a 24/7 medical guide, triaging patients to the appropriate services without offering dangerous automated diagnoses.

3. **Emergency-First Architecture:** Unlike Tabibi or Sihhatech, CityHealth features a dedicated "Emergency Mode", providing a specialized interactive map for instant geolocation of 24/7 services and blood donation centers.
4. **Absolute Data Integrity:** Every healthcare provider on CityHealth undergoes a strict administrative verification process, guaranteeing that citizens only access reliable, 100% accurate medical data.

To visually synthesize our competitive advantage, Table 2.1 presents a strict comparative analysis between CityHealth and the dominant solutions in the Algerian market.

Core Features	Google Maps	Facebook	Tabibi DZ	Dwaya	CityHealth
Doctor Appointment Booking	×	×	✓	×	✓
Verified Medical Profiles	×	×	✓	△	✓
Pharmacy & Lab Geolocation	✓	△	×	✓	✓
Dedicated 24/7 Emergency UI	×	△	×	×	✓
AI-Powered Patient Triage	×	×	×	×	✓
Multilingual (AR/FR/EN)	✓	✓	△	△	✓
Centralized Local Ecosystem	×	×	×	×	✓
Legend: ✓ Fully Supported △ Partially Supported / Unverified × Not Supported					

Table 2.1: Comparative Analysis of E-Health Solutions in Algeria

2.6 Conclusion of the Chapter

This chapter has established a comprehensive State of the Art by reviewing recent digital health literature in Algeria and critically analyzing the primary applications utilized by citizens today. The study clearly exposes a highly fragmented digital landscape lacking intelligent guidance and localized emergency management. By addressing these exact vulnerabilities through centralization, Artificial Intelligence, and rigorous data verification, CityHealth is perfectly positioned to capture the market and deliver unmatched value to the citizens of Sidi Bel Abbès. Having established our theoretical and competitive positioning, the following chapter will detail the specific methodologies, architectures, and technologies employed to build this innovative solution.

PART III

Modelling and Methodological Approach

CHAPTER 3

Preliminary Study and System Design

3.1 Chapter Introduction

This chapter is dedicated to the conceptual modeling and architectural design of the CityHealth platform. Before proceeding with the technical implementation, it is imperative to establish a clear blueprint of the system’s logic, data structures, and user interactions. To achieve this, we utilized globally recognized modeling standards, specifically the Merise methodology for database conception and the Unified Modeling Language (UML) for object-oriented system architecture and behavior mapping.

3.2 Merise Methodology

Merise is a comprehensive modeling methodology developed in France in the late 1970s and 1980s, which remains a standard for designing information systems, particularly in francophone academic and professional environments. It provides a structured approach to building an information system through separate abstraction levels: conceptual, logical, and physical. By separating the data processing from the data structure, Merise ensures a robust and scalable foundation for complex relational databases [6].

3.2.1 Conceptual Data Model (MCD)

The Conceptual Data Model, or *Modèle Conceptuel de Données (MCD)*, is the first and most abstract level of the Merise methodology. It represents the system’s data independently of any technical consideration, making it comprehensible to all stakeholders—whether developers, managers, or domain experts. The MCD uses **entities** (represented as rectangles), **associations** between entities (represented as ovals), and **cardinalities** to express the minimum and maximum number of relationships between entities [6]. It answers the question: *what data exists and how do things relate?*

3.3 Unified Modeling Language (UML)

The Unified Modeling Language (UML) is a standardized, general-purpose modeling language created by the Object Management Group (OMG). It was designed to provide a standard way to visualize the design of a system. UML standardizes several diagramming methods to represent both the structural and behavioral aspects of a software system [7]. For the CityHealth ecosystem, UML allows us to define how different actors interact with the web and mobile interfaces, as well as how the internal logic components communicate.

3.3.1 Use Case Diagram

A Use Case diagram is a behavioral UML diagram that summarizes the details of a system and the users (actors) interacting within it [8]. According to the OMG UML 2.5.1 specification, “a *UseCase* specifies a set of actions performed by its subjects, which yields an observable result that is of value for one or more Actors or other stakeholders of each subject” [9]. It is built from four main elements: the **system boundary**, **actors**, **use cases** (ovals), and **relationships** (association, include, extend, and generalization). For CityHealth, our primary actors include the **Citizen/Patient** (who searches for care and books appointments), the **Medical Professional** (who manages their clinic profile and appointment requests), the **Developer** (who consumes the API), and the **Administrator** (who verifies professional credentials and manages system integrity).

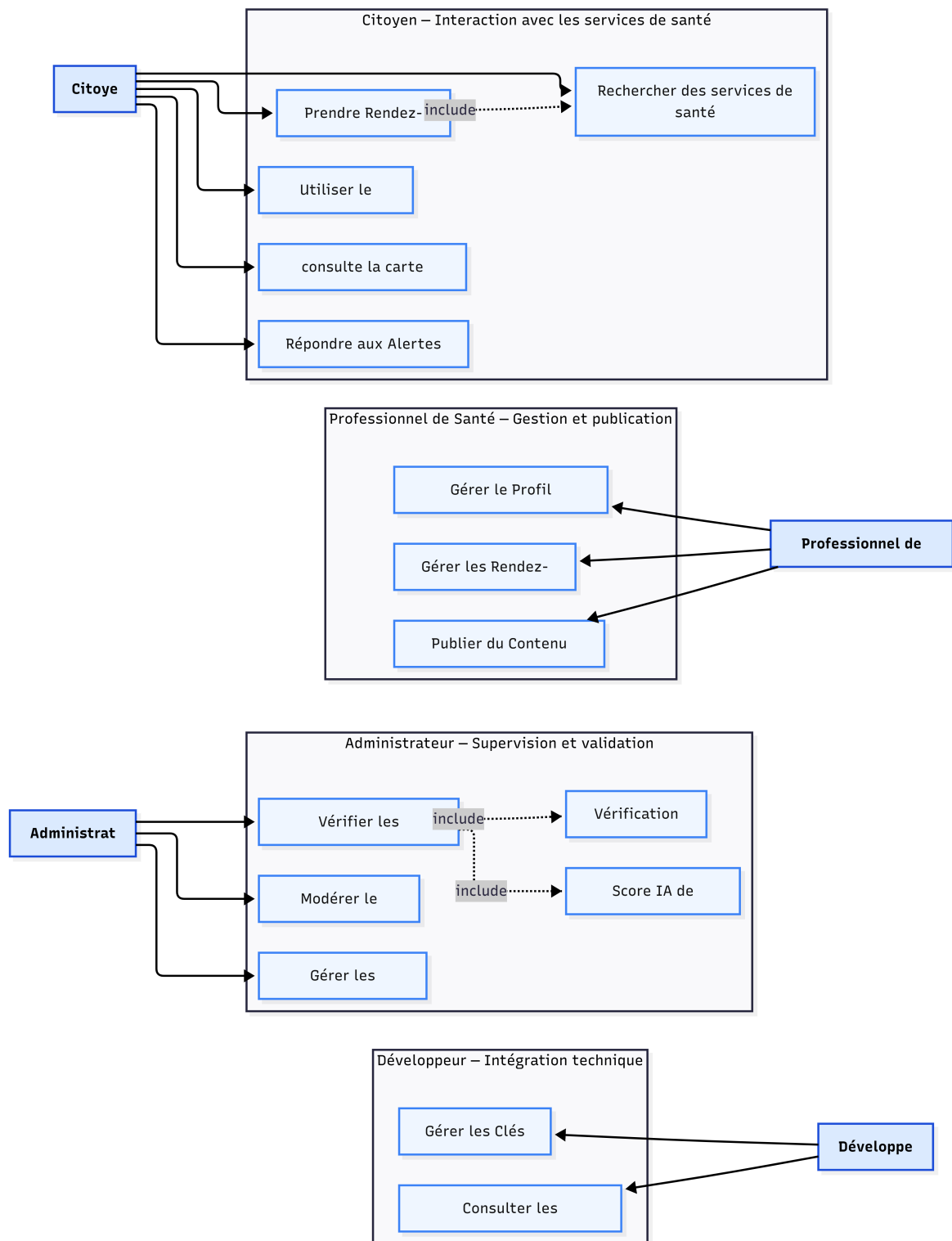


Figure 3.4: Global Use Case Diagram for the CityHealth Platform

3.3.2 Activity Diagram

An Activity diagram is a behavioral UML diagram that provides a view of the behavior of a system by describing the sequence of actions in a process [10]. It is essentially an advanced flowchart that models the flow from one activity to another, supporting

choice (diamond decision nodes), **iteration**, and **concurrency** (fork/join bars). Activity diagrams are constructed from action nodes, decision diamonds, a black circle for the initial node, and an encircled black circle for the final node [11]. In the context of CityHealth, the activity diagram captures the lifecycle of the medical article publication pipeline, illustrating conditional paths such as premium plan verification and content moderation.

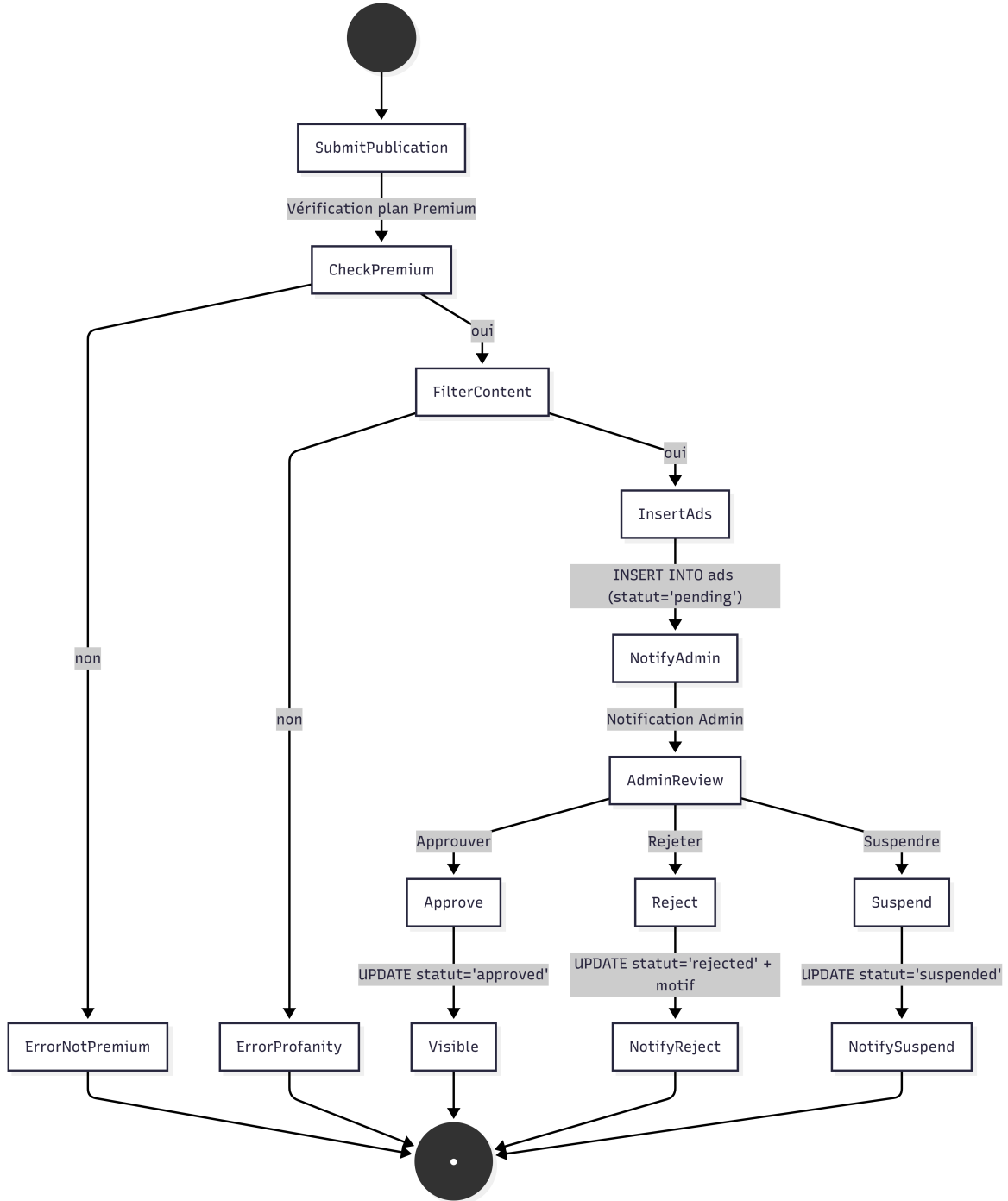


Figure 3.5: Activity Diagram — Medical Publication Submission Pipeline

3.3.3 General Class Diagram

A Class diagram is a structural UML diagram that models the static structure of a system [12]. It is the only UML diagram that maps directly to object-oriented programming languages, making it the blueprint of the system's architecture. A class diagram represents **classes** (with their attributes and methods), **interfaces**, and **relationships** such as associations, aggregations, compositions, dependencies, and inheritance (generalization) [13]. In the context of CityHealth, this diagram encapsulates the core logic, showcasing how a **Client** object interacts with a **RendezVous** object, which is inherently linked to a **Provider** object, all specializing the base **Utilisateur** class.

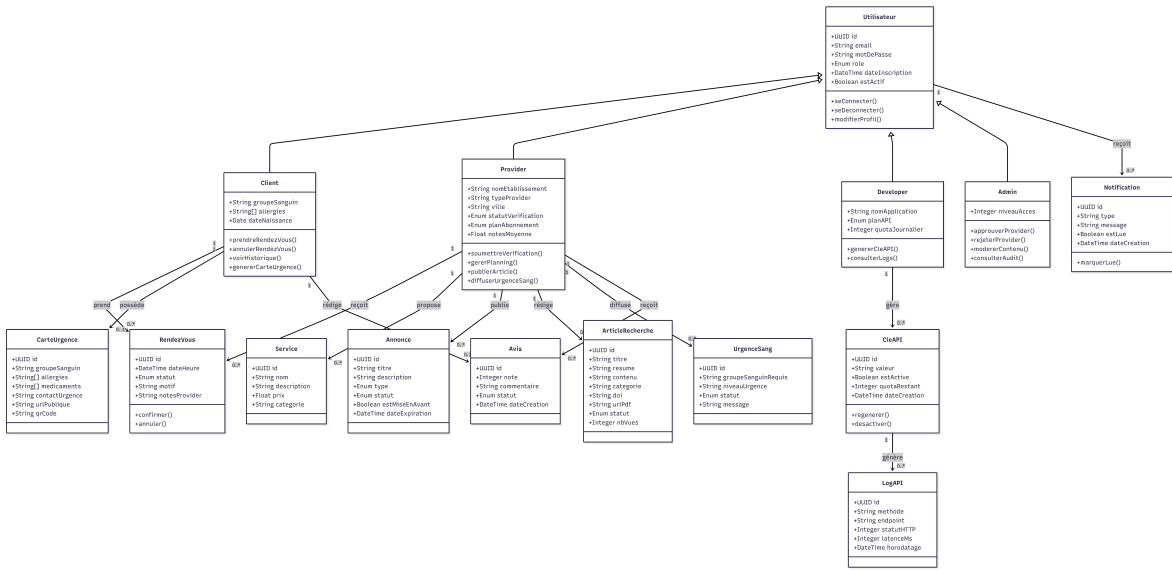


Figure 3.6: General Class Diagram of CityHealth

3.3.4 Sequence Diagrams

A Sequence diagram is a behavioral UML interaction diagram that represents the sequence of messages exchanged between objects during a specific scenario [14]. It shows parallel vertical **lifelines** representing the participants (actors and system components) and horizontal **arrows** for the messages exchanged, ordered chronologically from top to bottom [15]. Solid arrows represent synchronous calls while dashed arrows represent return messages or asynchronous responses. For CityHealth, we present three targeted sequence diagrams, each capturing a distinct critical workflow of the platform.

Authentication Sequence

This diagram models the complete authentication flow for the CityHealth platform. It details how the Frontend (React) invokes `supabase.auth.signInWithPassword`, how Supabase Auth verifies credentials against `auth.users` in PostgreSQL, retrieves the user role from the `user_roles` table, and finally returns a **JWT session token** to the frontend, which stores it in the Zustand state manager and redirects the user to the role-appropriate dashboard.

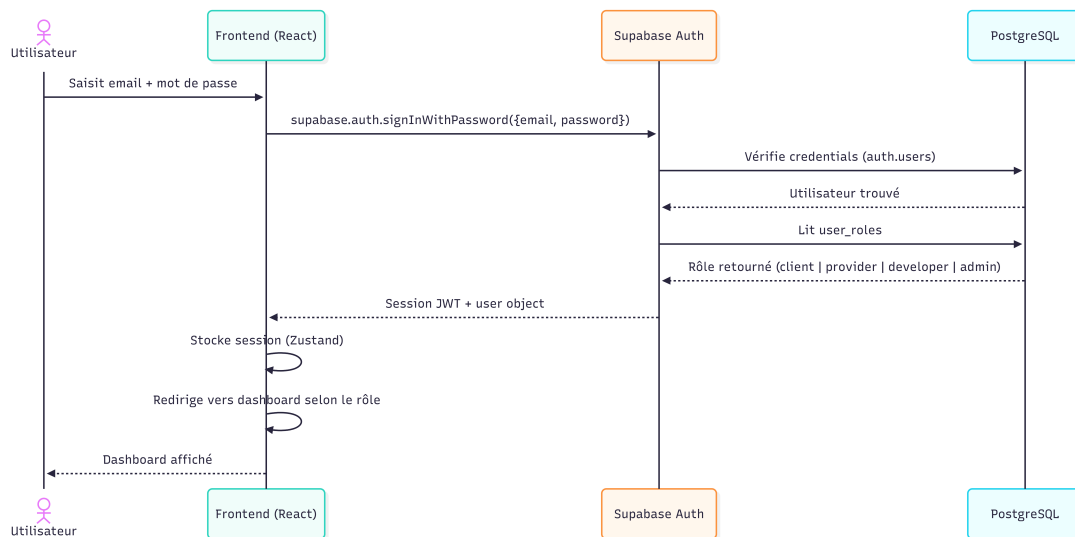


Figure 3.7: Sequence Diagram — User Authentication Flow

Appointment Lifecycle Sequence

This diagram captures the full lifecycle of a medical appointment on the CityHealth platform, from initial booking by a **Client** through provider confirmation. It covers three phases: (1) the **booking phase**, in which **appointmentService** inserts the record with `statut='pending'` and **notificationService** dispatches a confirmation email; (2) the **planning consultation phase**, in which the **Provider** queries pending appointments via a `SELECT WHERE provider_id = X`; and (3) the **confirmation phase**, in which the provider triggers an `UPDATE statut='confirmed'` followed by a real-time push notification to the client via **Supabase Realtime**.

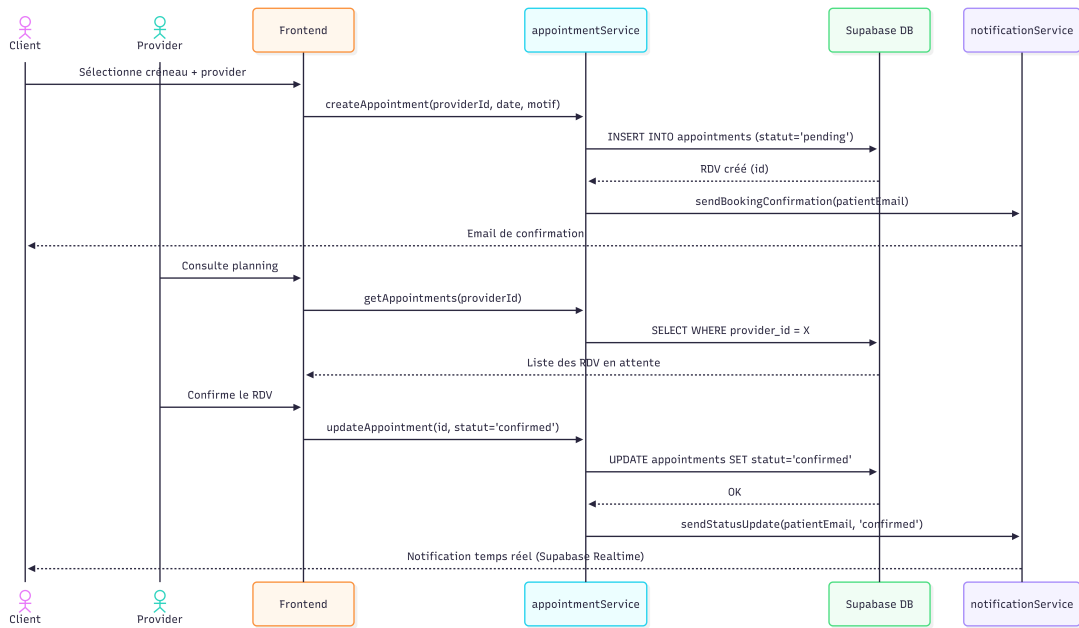


Figure 3.8: Sequence Diagram — Appointment Lifecycle

Provider Verification (OCR) Sequence

This diagram illustrates the provider credential verification pipeline, a key trust mechanism of the CityHealth platform. A **Provider** uploads a document (PDF or image), which is stored in **Supabase Storage**. The **ocrVerificationService** then applies **Tesseract.js** for multilingual text extraction (FR, EN, AR) followed by a **Levenshtein fuzzy match** to validate the extracted fields against expected data, producing a **VerificationResult** with a confidence score. The **verificationService** sets the provider status to **pending** in the database. An **Admin** then reviews the queue, using the OCR score as a decision aid, and upon approval triggers an **UPDATE verification_status='verified'**, an audit log entry, and a final notification to the provider.

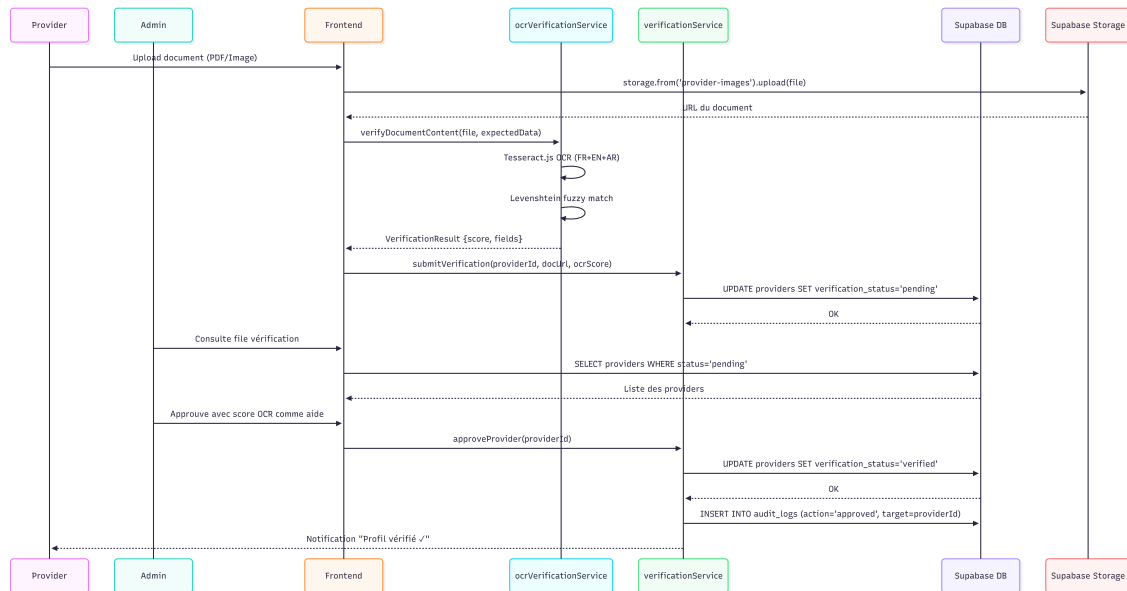


Figure 3.9: Sequence Diagram — Provider OCR Verification Flow

3.4 Technical Architecture and Security Model

3.4.1 Introduction

This section details the software architecture and the multi-layered cybersecurity mechanisms implemented within the CityHealth platform. Designed to ensure high performance, fault tolerance, and the absolute confidentiality of sensitive medical and practitioner data, the infrastructure relies on a modern, distributed, and highly secure serverless paradigm.

3.4.2 Global Technical Architecture

The CityHealth ecosystem is structured around a decoupled micro-modular architecture that promotes strict separation of concerns, high velocity, and horizontal scalability.

- **Frontend Application:** The client-side user interface is built using **React.js** and orchestrated by the **Vite** build tool. This technological combination ensures lightning-fast Hot Module Replacement (HMR) during development and yields optimized, lightweight production assets for an enhanced user experience.
- **Backend & API Layer:** The core API is instantly and securely generated via **PostgREST** (embedded natively within Supabase), eliminating the overhead and security surface area of a traditional monolithic Express.js server. Asynchronous processes, complex workflows, and Artificial Intelligence orchestrations are managed via serverless **Supabase Edge Functions** executed within the secure **Deno** runtime environment.
- **Database Layer:** The persistence layer and state management rely on a robust relational **PostgreSQL** database managed via Supabase, ensuring transactional integrity (ACID compliance) for critical medical records.
- **AI & Context Integration:** The platform's computational intelligence is driven by the **Google Gemini API (Gemini 1.5 Flash model)**, powering specialized medical chatbots for symptom triage and geospatial navigation guidance. Furthermore, the architecture implements the **Model Context Protocol (MCP)** to dynamically structure, isolate, and secure the context injected into the language models.

3.4.3 Defense-in-Depth Security Framework

The security paradigm of CityHealth strictly implements a *Defense-in-Depth* approach, assuming that no single layer is entirely impregnable. When data flows from the client

browser to the persistence layer, it encounters autonomous security boundaries at every stage. The frontend sanitizes inputs, the middleware layer filters traffic and prevents resource abuse, and the database engine enforces cryptographic row-level authorization before any read or write operation is committed.

Frontend Defense Mechanisms (React / Vite)

The first defensive layer operates entirely within the user’s browser runtime environment:

- **Secure Environment Variable Injection:** The compilation pipeline guarantees that no critical infrastructure secrets (such as the private Gemini API keys or master database administrative keys) are ever exposed to the client side. Only the public, unprivileged Supabase anonymous key (*anon key*) required for basic routing is distributed.
- **Native Cross-Site Scripting (XSS) Prevention:** By leveraging React’s data-binding architecture, the application inherently mitigates XSS vulnerabilities. All dynamic user inputs or server responses are automatically escaped before being rendered into the Document Object Model (DOM), neutralizing malicious script injection attempts.

Backend and API Security (Edge Functions & PostgREST)

Once a request leaves the client environment, it is intercepted by middleware components acting as strict rate-limiting and origin verifiers:

- **Sliding-Window Rate Limiting:** To protect the system against Distributed Denial of Service (DDoS) attacks and mitigate financial risks associated with AI API token consumption (Wallet Drain), an algorithmic sliding-window rate limiter is deployed. This mechanism restricts AI-related interactions to a maximum of **10 requests per minute per user**. Violations are instantly blocked at the edge with an HTTP 429 `Too Many Requests` status code.
- **Strict Cross-Origin Resource Sharing (CORS):** The backend explicitly intercepts HTTP `OPTIONS` preflight requests. It cryptographically enforces strict origin checks to guarantee that only the officially deployed CityHealth frontend domains can successfully invoke edge processing routines.

Database Security Layer (Supabase / PostgreSQL)

The database serves as the final and absolute validator of cryptographic authorization:

- **Row Level Security (RLS):** Every relational table handling sensitive data (including `ads`, `reviews`, and `quotes`) is locked down with strict RLS policies. Data mutations (`INSERT`, `UPDATE`, `DELETE`) are restricted directly at the database engine level based on the authenticated session token via the `auth.uid()` function. This architecture eliminates Broken Object Level Authorization (BOLA/IDOR) vulnerabilities, ensuring users can only manipulate records they explicitly own.
- **SQL Injection (SQLi) Elimination:** Interaction with the database layer is mediated exclusively through parameter-driven queries generated by the Supabase SDK. Raw string concatenation is completely avoided, treating all user-supplied parameters strictly as literal data values rather than executable SQL statements.

Cloud Storage Security (Storage Buckets)

Unstructured data files, such as medical provider credentials and practitioner profile documentation, are protected against broad reconnaissance:

- **Directory Scraping Deactivation:** While public storage buckets allow the downloading of legitimate individual assets via direct tokenized URLs, bulk directory enumeration is restricted. The directory listing routine (`.list()`) is completely blocked for unauthenticated or unauthorized scopes, preventing malicious actors from harvesting or scraping the data directory of medical practitioners globally.

3.5 Chapter Conclusion

This chapter presented the complete modeling foundation of the CityHealth platform. Through the Merise methodology, three successive levels of data abstraction were produced: the MCD captures the conceptual structure of the information domain, the MLD translates it into a relational schema, and the MPD specifies its physical implementation in PostgreSQL via Supabase. These were complemented by a suite of UML diagrams—a Use Case diagram defining the functional perimeter for all four actor roles, an Activity diagram tracing the content publication pipeline, a General Class diagram representing the object-oriented architecture, and three Sequence diagrams detailing the critical runtime flows of authentication, appointment management, and OCR-based provider verification. Together, these artifacts provide a rigorous and comprehensive blueprint for the implementation phase described in the following chapter.

PART IV

Scientific Implementation and Results

CHAPTER 4

Scientific Implementation and Results

4.1 Introduction

This chapter is dedicated to the technical realization of the CityHealth project. After modeling our architecture in the previous chapter, we present here the collaborative development environment established, the steps for building our Minimum Viable Product (MVP), and the implementation of our processing algorithms (OCR, moderation). Finally, we will present the results from our testing protocols (unit, E2E, and performance) to validate the platform’s functionality under real-world conditions.

4.2 Work Environment and Tools

Given that the development was carried out as a pair, setting up a standardized work environment and robust collaboration tools across our two machines was paramount to ensure seamless integration.

Table 4.1: *Hardware specifications of the collaborative development environment*

Component	Workstation 1 (Naimi)	Workstation 2 (Belyagoubi)
Operating System	Windows 11	Windows 11
RAM	16 GB	16 GB
CPU	Intel Core i7-10750H	AMD Ryzen 5 5600U (2.30 GHz)
GPU	NVIDIA GeForce GTX 1650 Ti (4 GB)	AMD Radeon Graphics

4.3 Prototype (MVP) Realisation

The realization of the MVP revolved around several interconnected deliverables forming the complete CityHealth ecosystem. To demonstrate the comprehensiveness of the platform, the following subsections detail the implemented interfaces.

4.3.1 Core Platform & Landing Interfaces

The public-facing interfaces were designed with a focus on accessibility, speed, and clear user journeys.



Figure 4.1: The CityHealth main landing page, featuring an intuitive, centralized search engine for medical professionals in Sidi Bel Abbès, complemented by quick-access specialty tags and real-time platform statistics.



Figure 4.2: A visual guide outlining the seamless user journey on the platform, illustrating the three-step process for patients: searching verified databases, comparing profiles, and booking appointments.

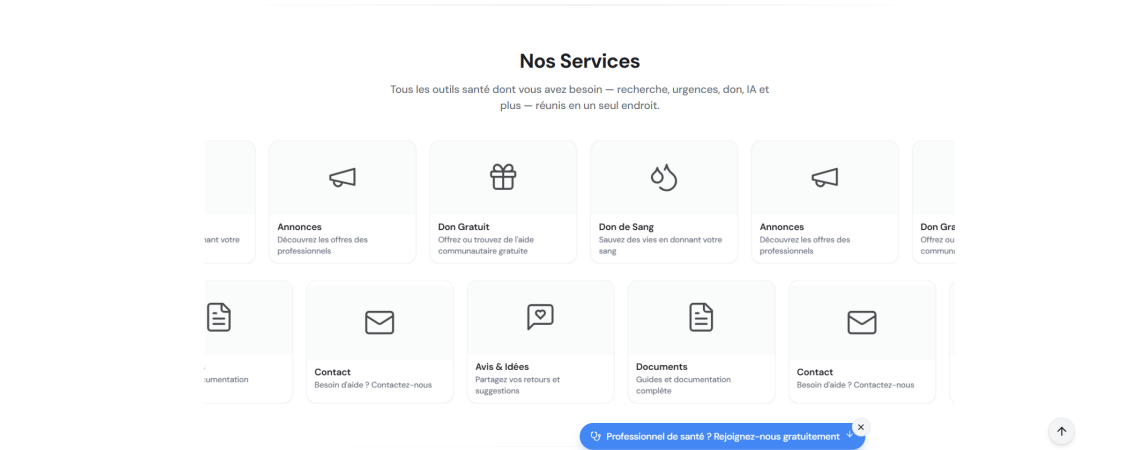


Figure 4.3: An interactive interface showcasing the comprehensive suite of CityHealth services, extending beyond medical booking to include critical community features such as blood donation alerts and medical equipment sharing.

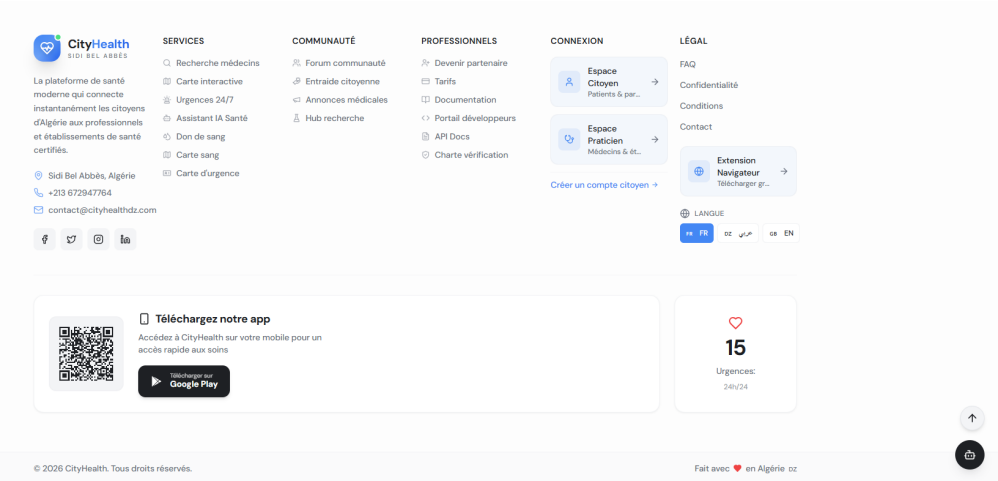


Figure 4.4: The comprehensive platform footer, highlighting complete multilingual support (i18n), seamless mobile application integration, and dynamic real-time platform statistics.

4.3.2 Authentication & User Onboarding

Security and strict Role-Based Access Control (RBAC) dictate distinct entry paths for patients and healthcare providers.

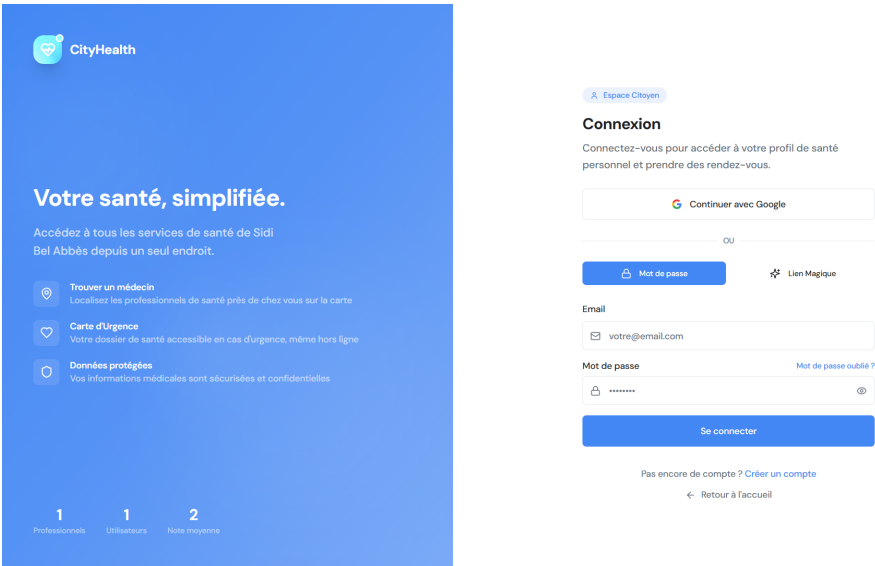


Figure 4.5: The secure citizen authentication portal, demonstrating the integration of modern access methods including OAuth providers and passwordless Magic Links via Supabase Auth.

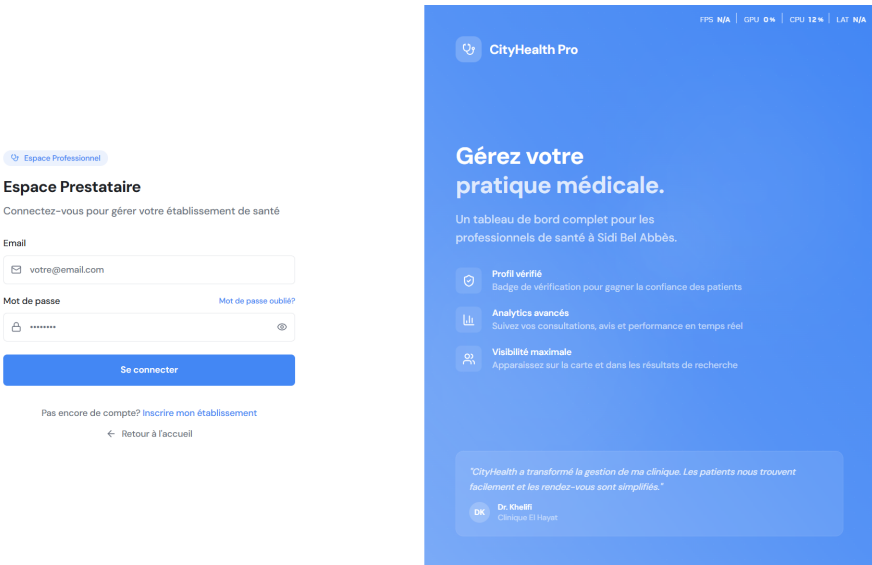


Figure 4.6: The dedicated authentication gateway for healthcare professionals, enforcing strict Role-Based Access Control (RBAC) and tailored specifically for medical providers.



Figure 4.7: The dedicated onboarding portal for healthcare professionals, detailing the streamlined registration, profile completion, and strict verification process required to join the medical directory.

4.3.3 User Dashboards & Provider Ecosystem

Once authenticated, users access specialized dashboards tailored to their specific health-care needs or professional management requirements.

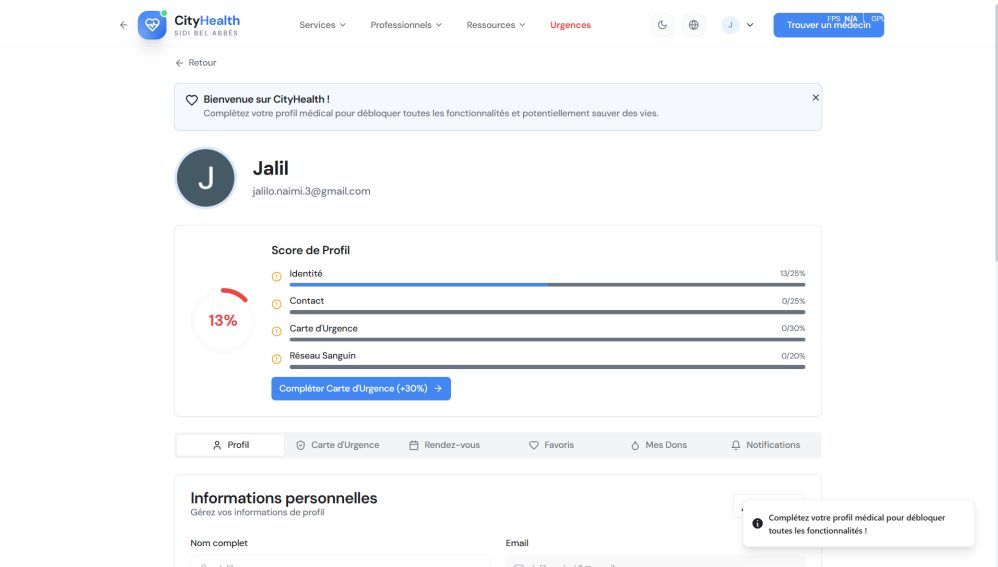


Figure 4.8: The centralized patient dashboard, featuring a dynamic profile completion tracker, secure appointment management, and quick access to emergency and donation records.

← Retour

Inscription Professionnel

Rejoignez CityHealth et connectez-vous avec des milliers de patients

14%

Score de profil

Complétez votre identité pour atteindre 40%

Non visible

1

Identité (0/20)

2

Institution (0/20)

3

Visibilité (12/30)

4

Bonus (2/30)

Brouillon sauvegardé 19:09

Étape 1 / 6

1

Identité

2

Établissement

3

Localisation

4

Services

5

Profil

6

Aperçu

Sauvegarder et quitter

Bienvenue sur CityHealth Pro

Rejoignez notre réseau de professionnels de santé et développez votre visibilité auprès de milliers de patients

Données sécurisées Rejoindre l'annuaire de la santé numérique

Figure 4.9: The multi-step provider onboarding interface, designed to securely gather identity, institutional, and operational data while tracking profile completeness prior to administrator verification.

CityHealth
SIDI BEL ABBÉS

Services

Professionnels

Ressources

Urgences

Connexion

Trouver un médecin

Test Clinique

PREMIUM

Clinique

2.0 3 avis

Accessible PMR

Prendre rendez-vous

À propos

Clinique moderne avec équipement de pointe

Sidi Bel Abbès

Langues: fr

Avis des patients

2.0

Basé sur 3 avis

5 4 3 2 1

0 1 0 0 2

Contact

123 Test Street

0555555555

Appeler

Ouvert maintenant

Horaires

Lun 08:00 - 18:00

Mar 08:00 - 18:00

Mer 08:00 - 18:00

Jeu 08:00 - 18:00

Ven 08:00 - 12:00

Sam 09:00 - 12:00

Dim Fermé

Prochains créneaux disponibles

Figure 4.10: The public-facing profile of a verified healthcare provider, displaying critical operational details, facility information, and an aggregated patient review score.

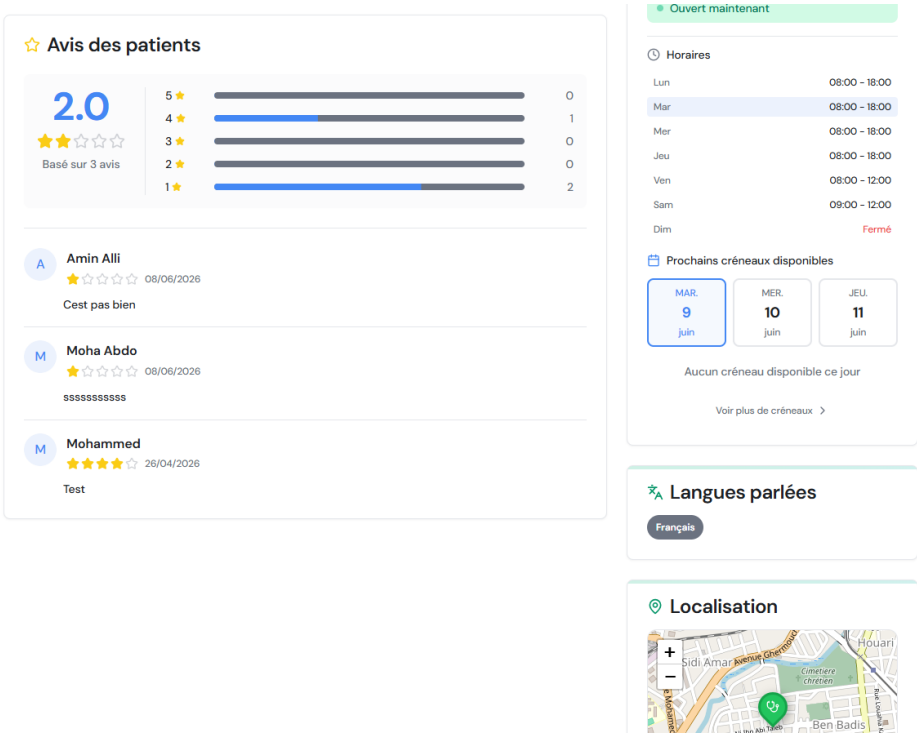


Figure 4.11: The lower section of the provider profile, highlighting the transparent patient review system, available consultation slots, and a localized mini-map for precise navigation.

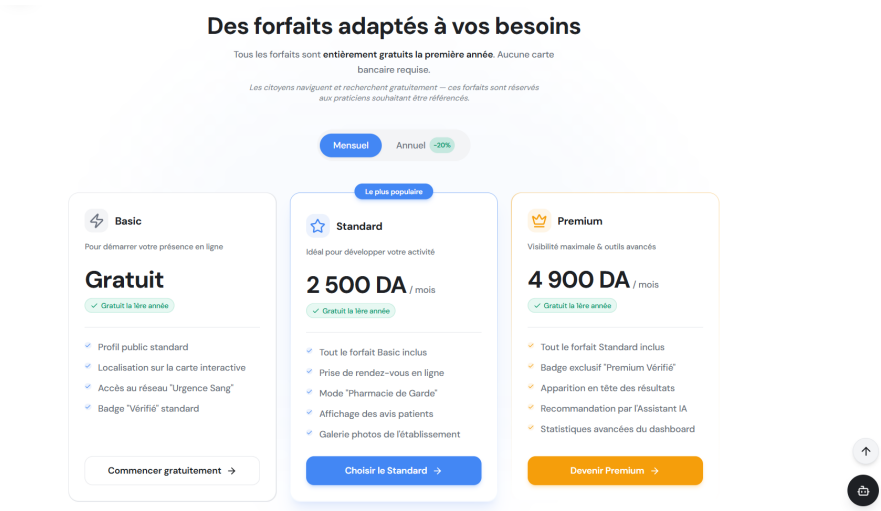


Figure 4.12: The tiered subscription model interface for healthcare providers, clearly delineating the functionalities of the Basic, Standard, and Premium plans, including advanced dashboard analytics and AI-driven visibility.

4.3.4 Geospatial & Search Capabilities

Location-based services are a core pillar of the platform, built using Leaflet and Mapbox integrations.

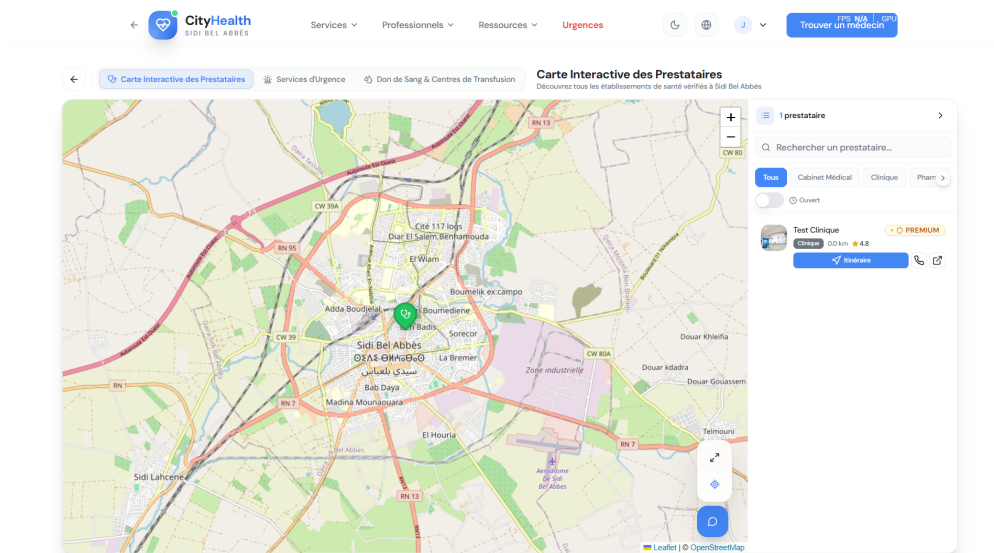


Figure 4.13: The interactive geolocation map interface, utilizing Leaflet to display verified healthcare providers in Sidi Bel Abbès, complete with dynamic category filtering and real-time operational status.

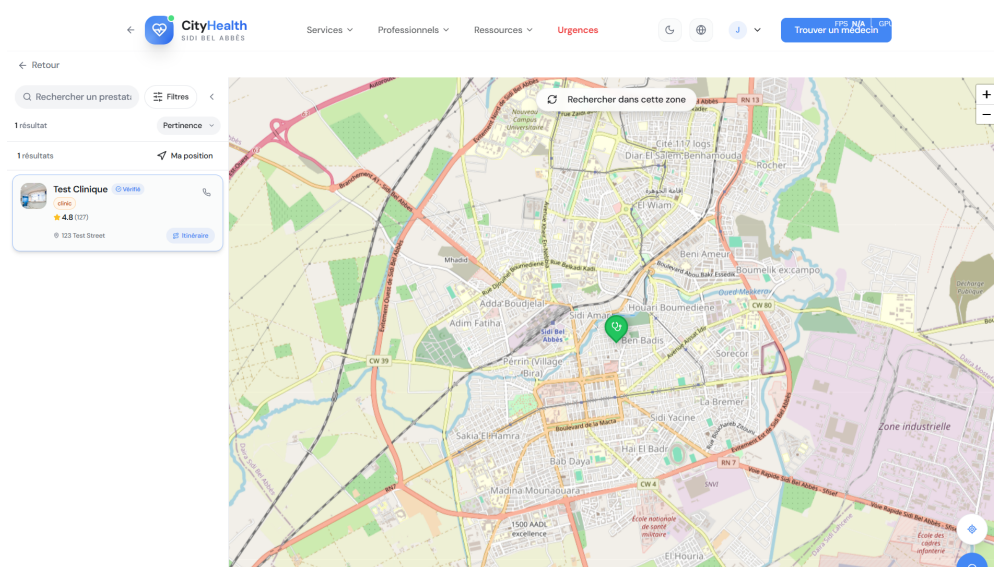


Figure 4.14: The geospatial search interface, allowing patients to dynamically locate nearby clinics and healthcare professionals using interactive map markers and advanced filtering criteria.

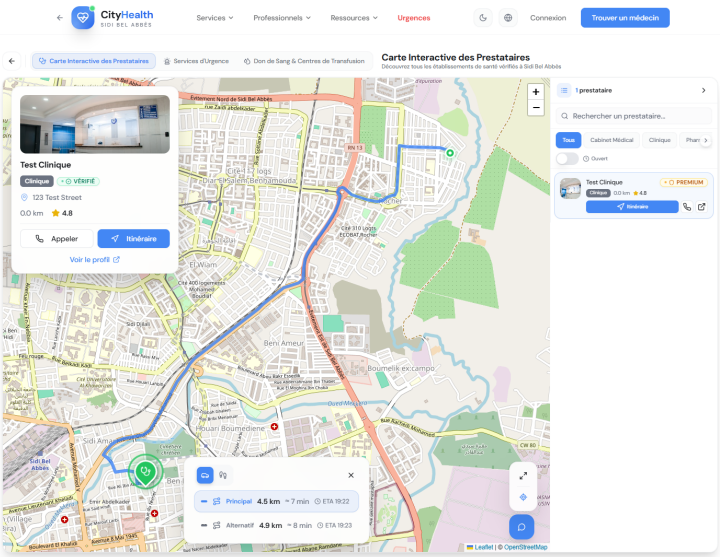


Figure 4.15: Advanced geolocation and routing integration within the interactive map, providing patients with real-time distance calculations and optimal itineraries to healthcare facilities.

4.3.5 Emergency Response & Community Mutual Aid

CityHealth distinguishes itself through real-time community solidarity features powered by Supabase WebSockets.

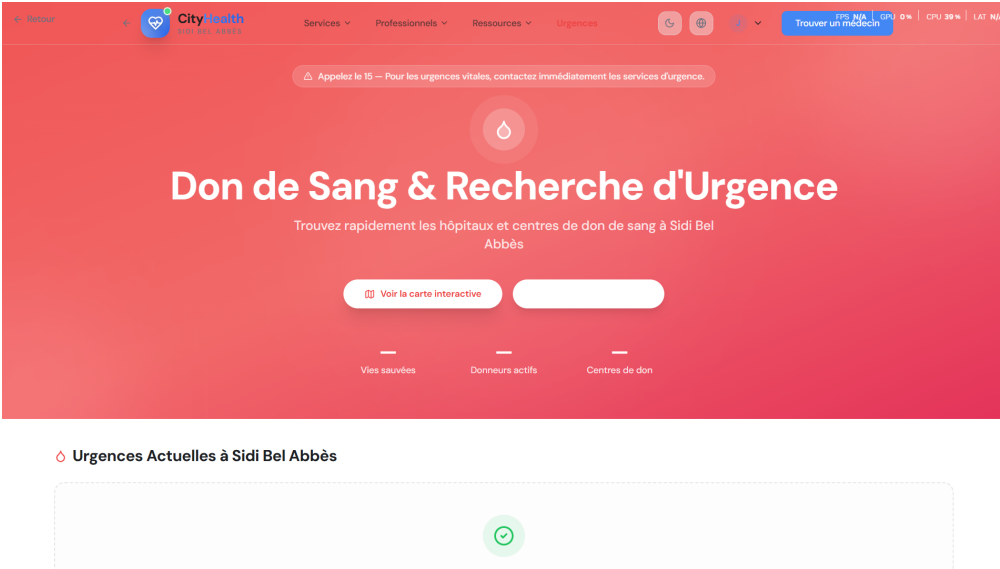


Figure 4.16: The dedicated blood donation and emergency response dashboard, designed to broadcast urgent blood requirements in real-time and display critical community engagement statistics.



Figure 4.17: The personalized Medical Emergency Card, generating a unique QR code that grants first responders instant access to critical patient data, with built-in functionality for local PDF exportation.

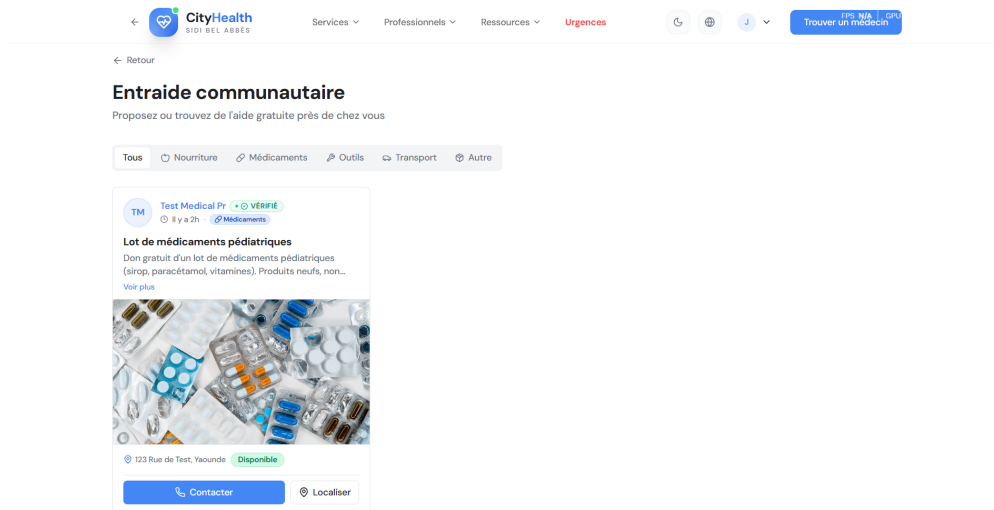


Figure 4.18: The community mutual aid and donation board, allowing verified users to locally offer or request medical supplies, medications, and health-related assistance.

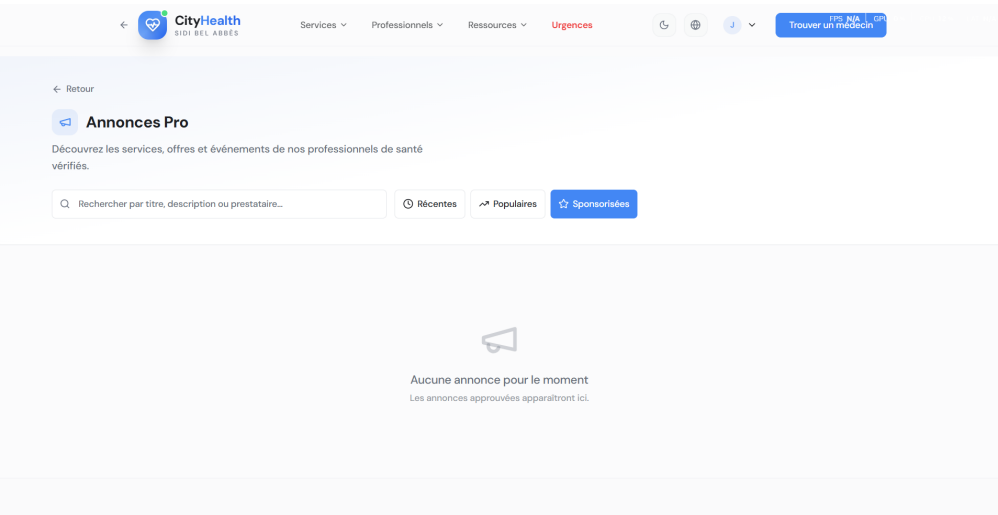


Figure 4.19: *The Professional Announcements board, providing a moderated communication channel for verified healthcare providers to publish medical news, clinic updates, and sponsored health services.*

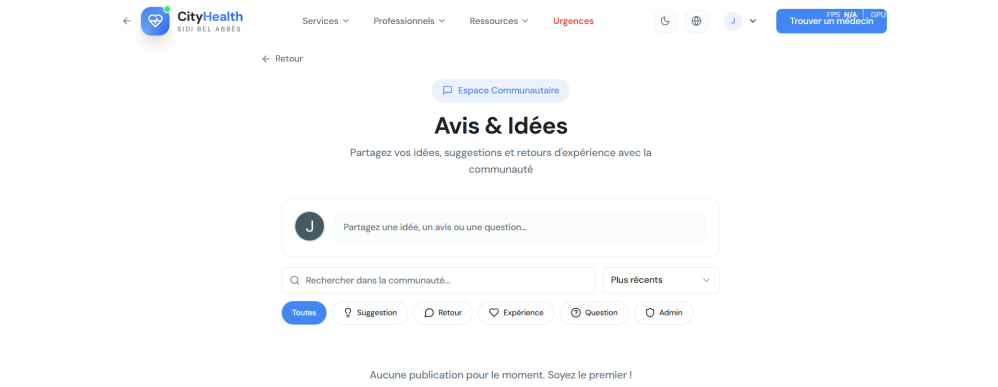


Figure 4.20: *The interactive community forum, featuring categorized interaction streams (such as suggestions, patient experiences, and questions) designed to foster transparent engagement between citizens and the platform administration.*

4.3.6 AI Integration & Medical Research

The integration of Artificial Intelligence and academic resources elevates the platform's utility.

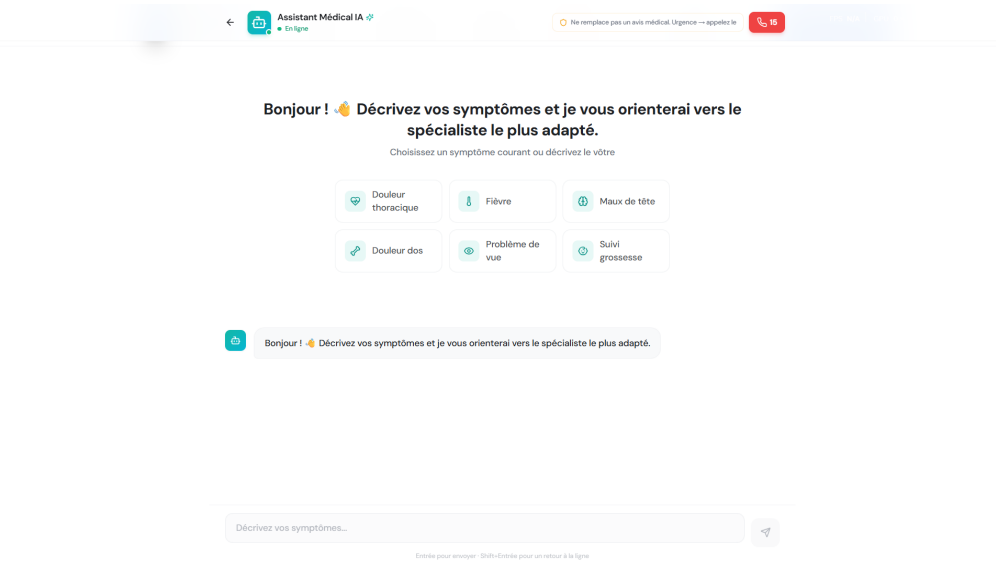


Figure 4.21: The AI-powered medical assistant interface, utilizing large language models (LLMs) to perform initial symptom triage and intelligently orient patients to the most appropriate healthcare specialists.

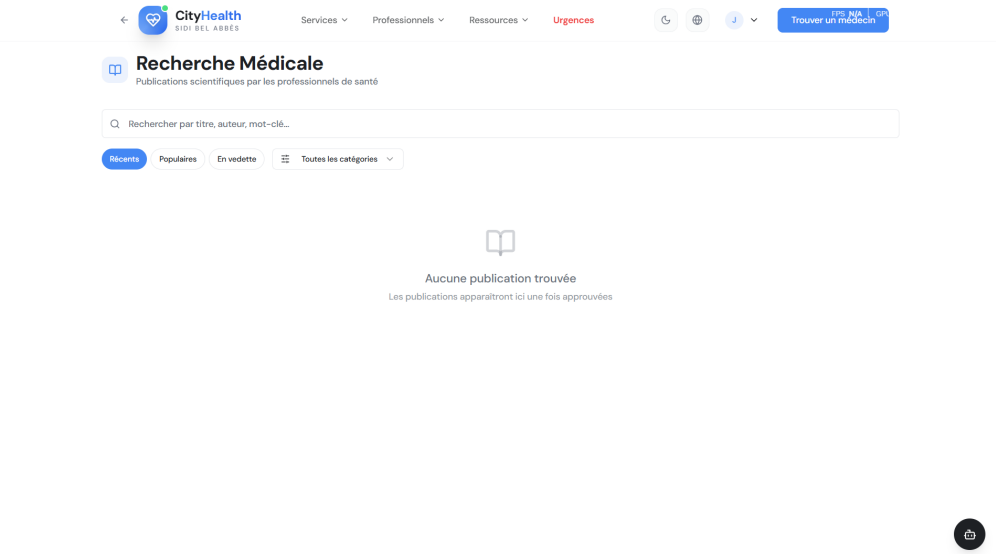


Figure 4.22: The dedicated Medical Research Hub, functioning as an academic repository where professionals can publish, search, and access verified scientific articles and clinical findings.

4.3.7 Developer API & Support Infrastructure

To ensure scalability and third-party integrations, comprehensive B2B tools were developed.

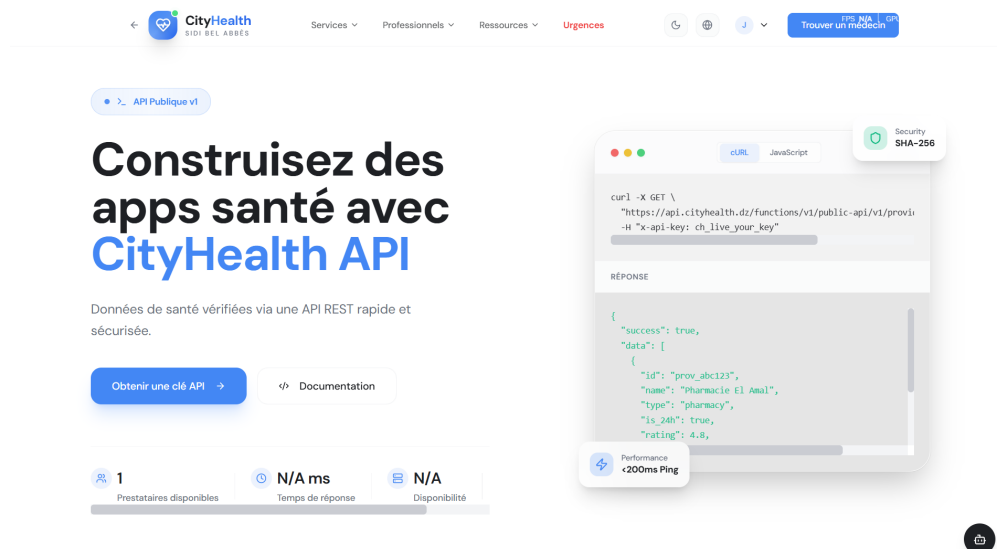


Figure 4.23: The CityHealth Public API developer portal, illustrating secure integration endpoints, SHA-256 API key authentication, and expected JSON payload structures for third-party health applications.

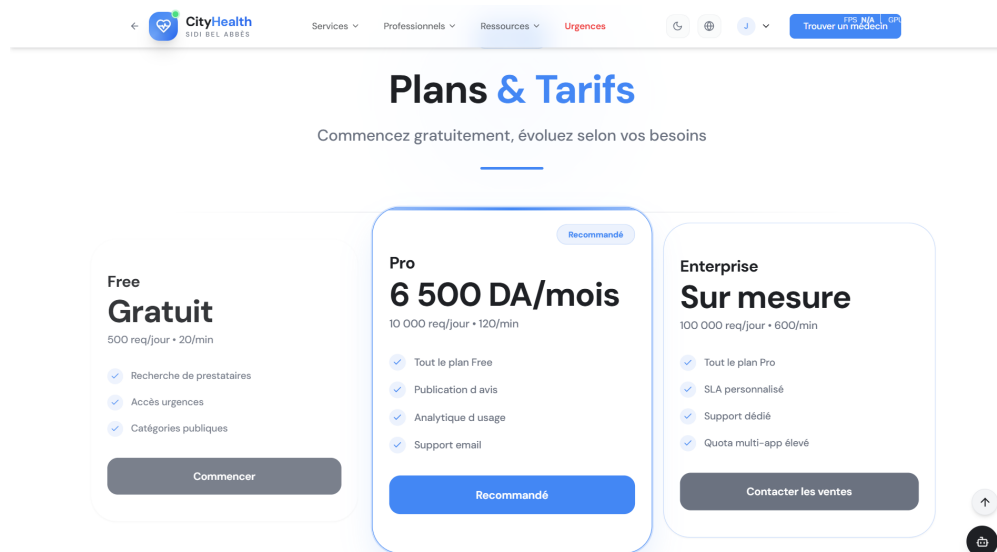


Figure 4.24: The tiered API quota management and pricing interface, detailing the specific rate limits (requests per day) and monetization strategy implemented for third-party developer access.



Figure 4.25: The comprehensive CityHealth documentation hub, featuring structured navigation for different user roles and an integrated AI assistant widget to provide contextual, real-time guidance.

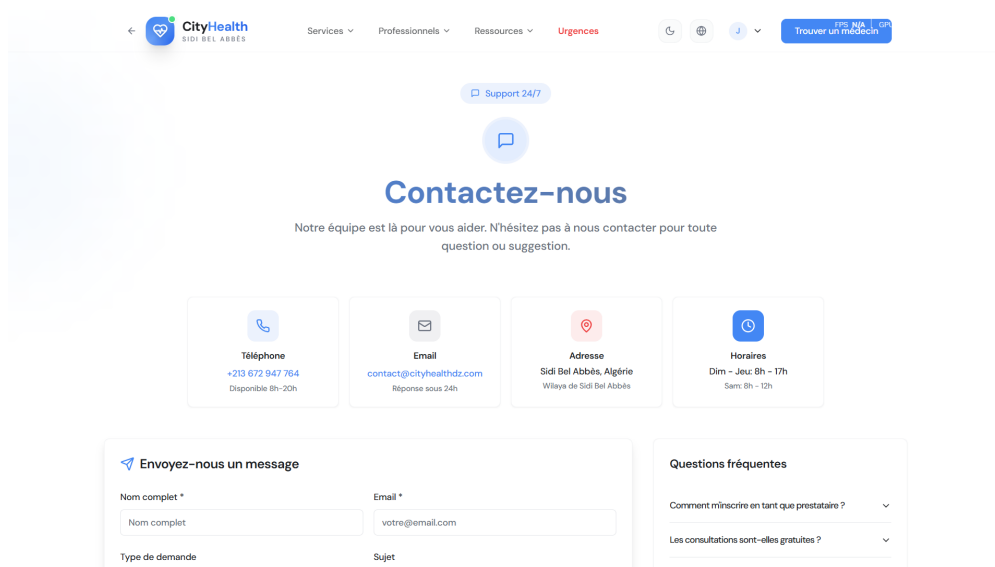


Figure 4.26: The centralized user support and contact portal, integrating a secure, server-less messaging form and a dynamic Frequently Asked Questions (FAQ) module to streamline platform assistance.

4.3.8 Mobile Application

To maximize accessibility and ensure that healthcare services remain available to citizens on the move, a dedicated mobile application was developed. The mobile experience mirrors the core features of the web platform while leveraging native device capabilities (push notifications, geolocation, and QR scanning) to deliver a streamlined, user-centric interface.

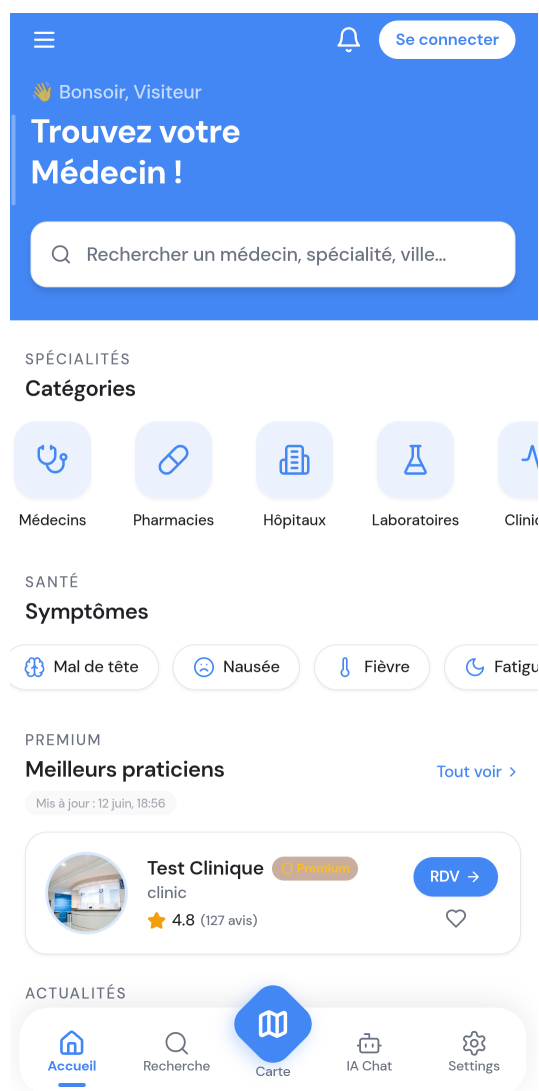


Figure 4.27: Primary mobile home screen introducing the CityHealth ecosystem, designed to make access to core services and navigation between features immediately understandable for new users.

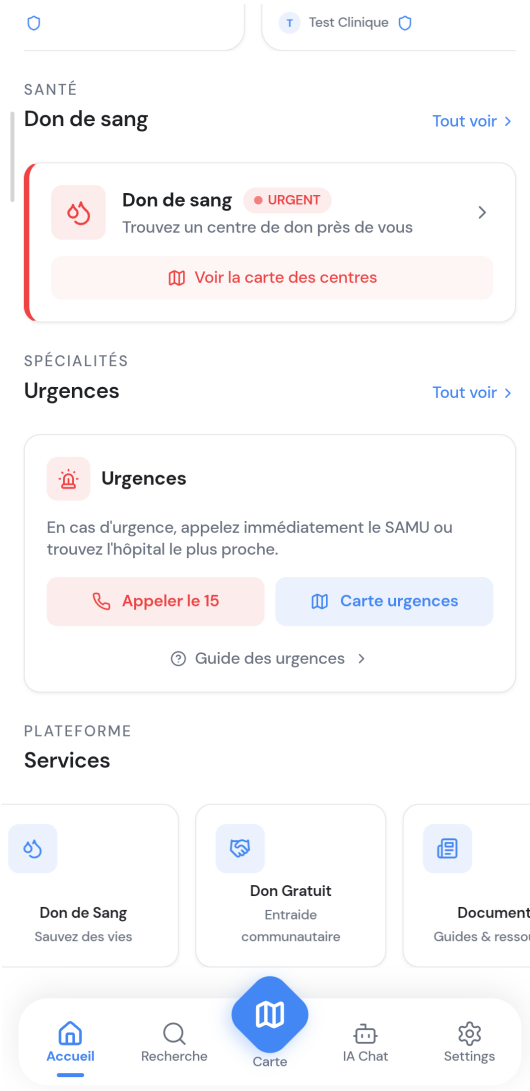


Figure 4.28: Secondary home view highlighting the main care pathways and shortcuts, simplifying entry to blood donation, emergency care, and service discovery directly from the landing screen.

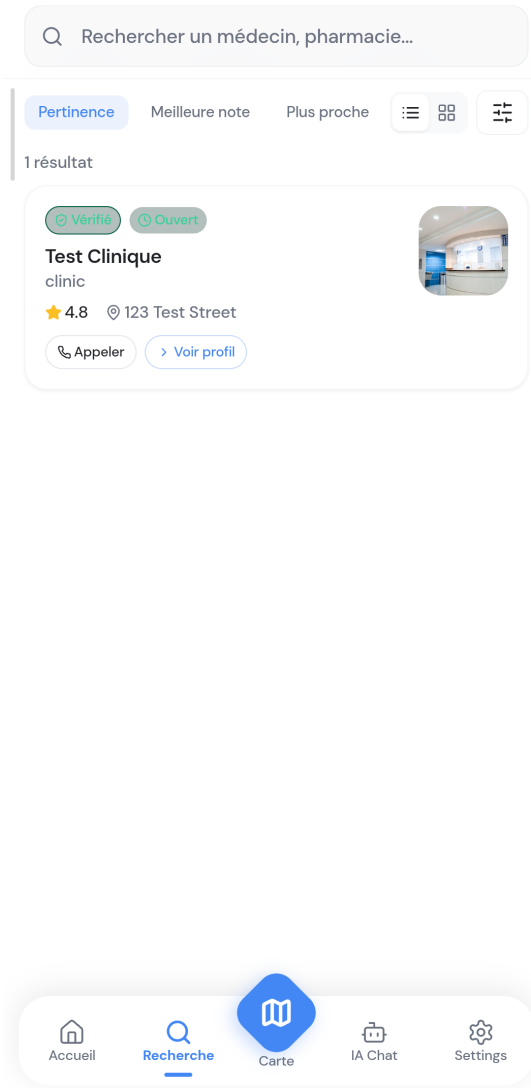


Figure 4.29: Unified search interface for providers and clinics, enabling citizens to quickly look up medical professionals, facilities, and health services using a familiar search experience.

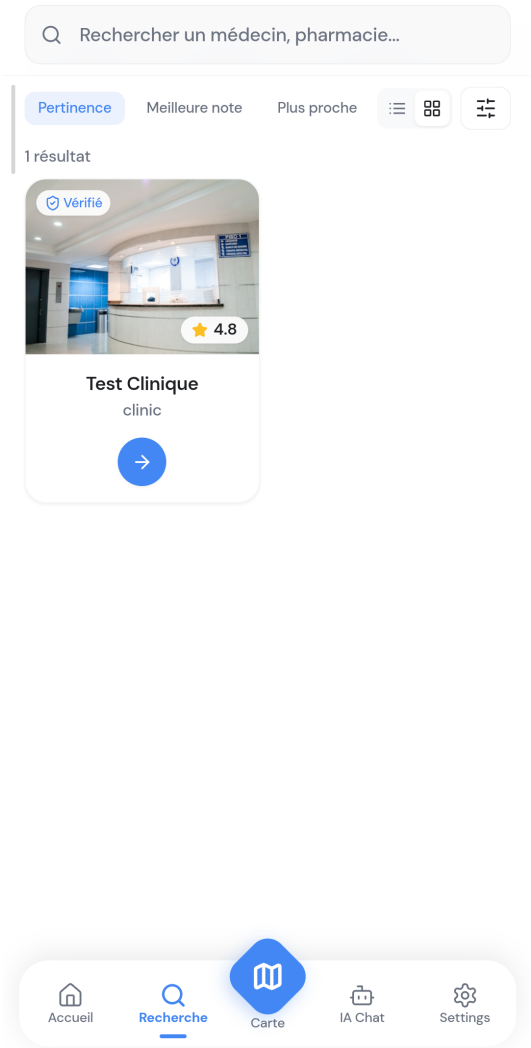


Figure 4.30: *second unified search interface for providers and clinics, enabling citizens to quickly look up medical professionals, facilities, and health services using a familiar search experience.*

The image shows a mobile application interface for finding medical services. At the top, there is a search bar with the placeholder text "Rechercher un médecin, pharmacie...". Below the search bar, there are four tabs: "Pertinence", "Meilleure note", "Plus proche", and a button with a grid icon. Below these tabs, it says "1 résultat".

The main part of the image is an "Filtres avancés" (Advanced filters) panel, which is a modal window with a close button (X) in the top right corner. The panel contains several filter sections:

- Disponibilité** (Availability): Four buttons: "Tous" (selected), "Ouvert maintenant", "Aujourd'hui", and "Cette semaine".
- Note minimum** (Minimum rating): Five star icons, with the last one selected, and the text "Toutes les notes".
- Options**: Three checkboxes, all of which are unchecked:
 - ☐ Vérifiés uniquement (Verified only)
 - ☐ Services d'urgence (Emergency services)
 - ☐ Accessibilité fauteuil roulant (Wheelchair accessibility)
- Langues parlées** (Languages spoken): A globe icon followed by three buttons: "Français", "العربية" (Arabic), and "English".
- Distance maximale** (Maximum distance): A slider bar with a blue track and a white knob. The text "50 km" is at the right end of the slider.

At the bottom of the panel, there are two buttons: "Réinitialiser" (Reset) and "Appliquer (1)" (Apply (1)).

Figure 4.31: Advanced filtering panel, allowing refinement of search results by availability, rating, spoken languages, and accessibility options to better match patient constraints.

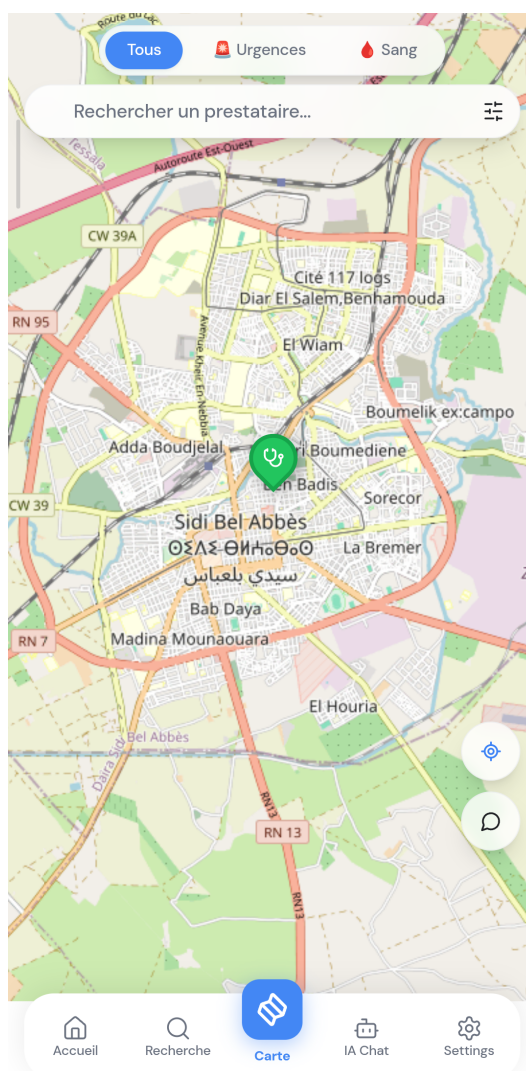


Figure 4.32: Interactive map view displaying healthcare providers, emergency services, and blood donation centers around Sidi Bel Abbès, with real-time category filters for rapid orientation.

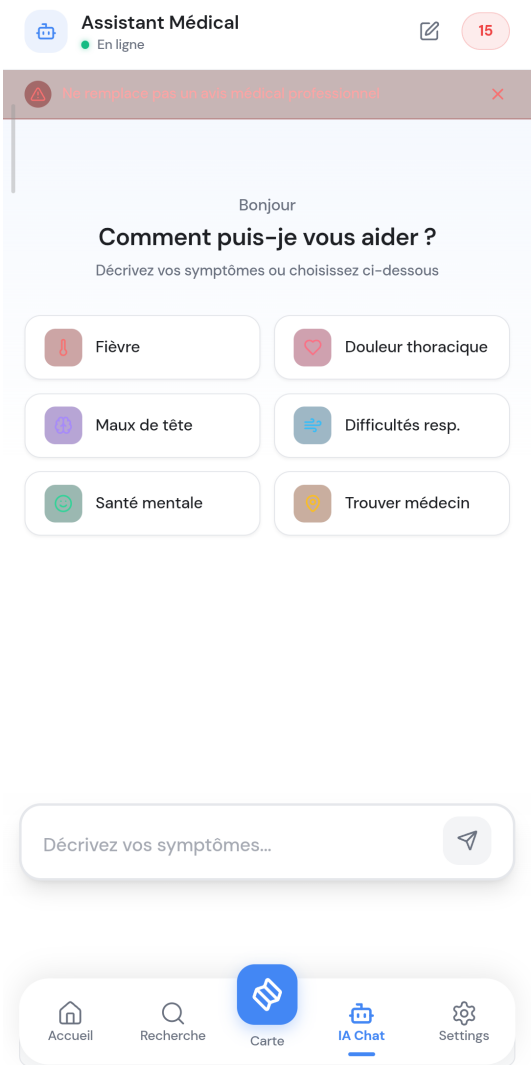


Figure 4.33: *Embedded AI medical assistant, offering a conversational interface for initial symptom triage and intelligent redirection towards the most relevant health services.*

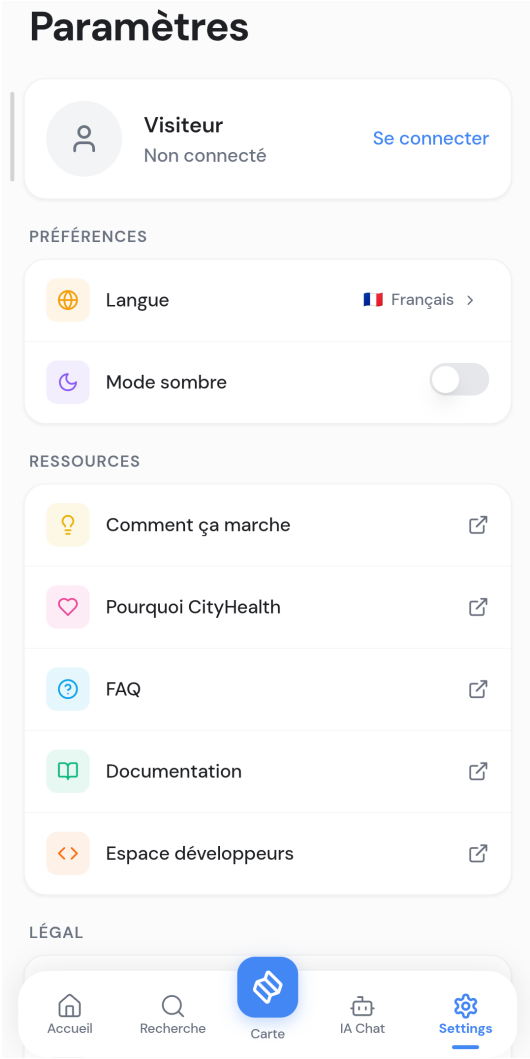


Figure 4.34: *Settings interface centralizing language selection, dark mode, access to documentation, FAQs, and the developer space, ensuring full control over the mobile experience.*



Figure 4.35: Dedicated blood donation module summarizing key statistics and urgent calls, encouraging citizens to participate in life-saving donation campaigns directly from their phone.



Figure 4.36: Mobile-to-web bridge, presenting services that are currently only available on the CityHealth web platform and redirecting users seamlessly via QR code.

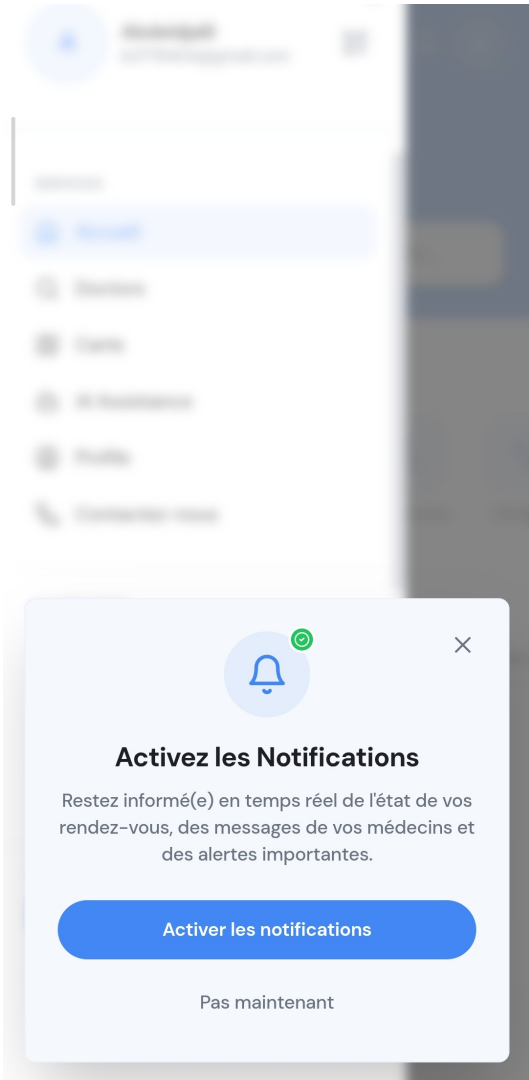


Figure 4.37: Notification activation flow, enabling users to opt in to real-time alerts for appointments, messages from healthcare providers, and critical platform notifications.

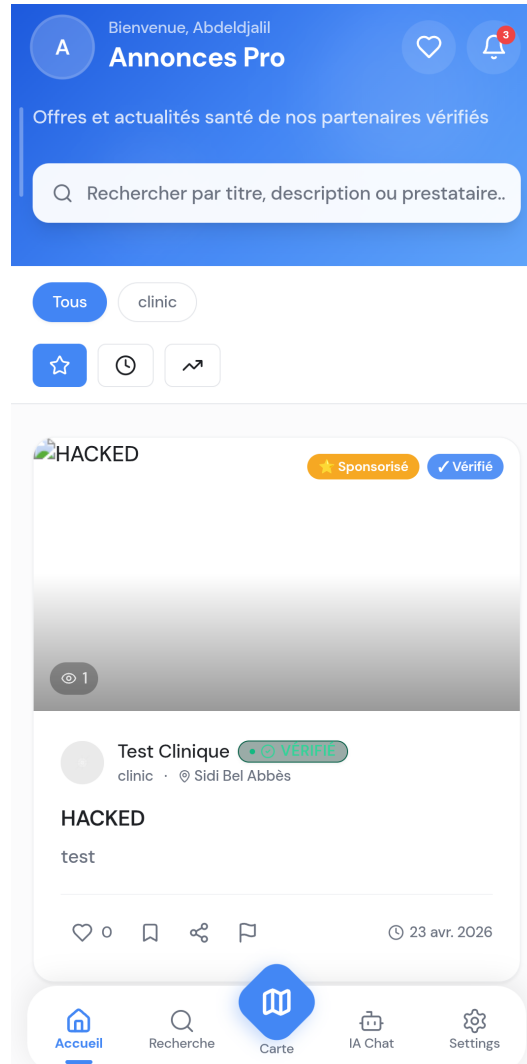


Figure 4.38: Professional announcements feed reserved for subscribed and verified providers, used to publish sponsored health offers, clinic updates, and public health campaigns to the mobile audience.

4.4 Tests, Experiments and Results

4.4.1 Test Protocol

The system was subjected to a rigorous testing pipeline across both of our development machines:

- **Unit Tests (Vitest):** Validation of critical utility functions (Levenshtein distance algorithm, content filtering logic).
- **End-to-End Tests (Playwright):** Simulation of critical user paths (full appointment booking flow, document upload, and admin validation).
- **API Latency Tests:** Measuring the response times of *Edge Functions* and the real-time broadcasting system.

4.4.2 Results Obtained

Table 4.2: *Summary of prototype performance*

Metric	Target (Theory)	Result (MVP)
OCR extraction accuracy	> 85%	~ 92%
Public API Latency	< 200 ms	120 ms (average)
Edge Function Execution Time	< 1500 ms	800 ms (average)
Code coverage (Vitest)	> 70%	78%

4.5 Discussion of Results

The results obtained confirm the technical viability of the chosen serverless architecture. The prototype runs smoothly, even during heavy asynchronous workflows (such as OCR extraction). However, the experimentation raised a few limitations: the quality of optical recognition drops significantly if the doctor photographs their diploma in poor lighting conditions. Integrating client-side image preprocessing (cropping, contrast adjustment) would be a highly relevant future improvement. Furthermore, the implementation of the MCP standard proved particularly effective, demonstrating that CityHealth is fully prepared to be integrated into the rapidly emerging ecosystem of artificial intelligence agents.

4.6 Conclusion of Part I

This first part allowed us to validate our technical choices through the realization of a robust, highly performant MVP strongly oriented towards modern innovation (AI, real-time sync, MCP). The architectural design, coupled with our experimental results, clearly demonstrates the technical feasibility of CityHealth. The following part (Part II) will shift focus to the managerial and entrepreneurial aspects, analyzing the economic viability of this solution as a tech startup (Business Plan, market study, and monetization strategy).

PART V

Economic Dimension — Startup Project

CHAPTER 5

Startup Business Model and Financial Study

5.1 Introduction to Part II

The transition from a scientific prototype to a viable socio-economic product requires a robust entrepreneurial strategy. Part I of this thesis demonstrated the technical feasibility and architectural design of CityHealth. This second part focuses entirely on the economic dimension of the project. CityHealth is not merely an academic exercise; it is designed as an innovative Algerian startup aimed at democratizing access to urban healthcare in Sidi Bel Abbès. Its primary mission is to eradicate the local medical information asymmetry through a centralized digital ecosystem.

To ensure the sustainability and scalability of this mission, a clear commercial strategy is required. This chapter details the Business Model Canvas (BMC) that dictates how CityHealth creates, delivers, and captures value, followed by a comprehensive financial study evaluating the initial investments, projected revenues, and overall economic viability of the startup.

5.2 Business Model Canvas

The Business Model Canvas (BMC) is a strategic management template used for developing new business models. It visually charts elements describing CityHealth's value proposition, infrastructure, customers, and finances.

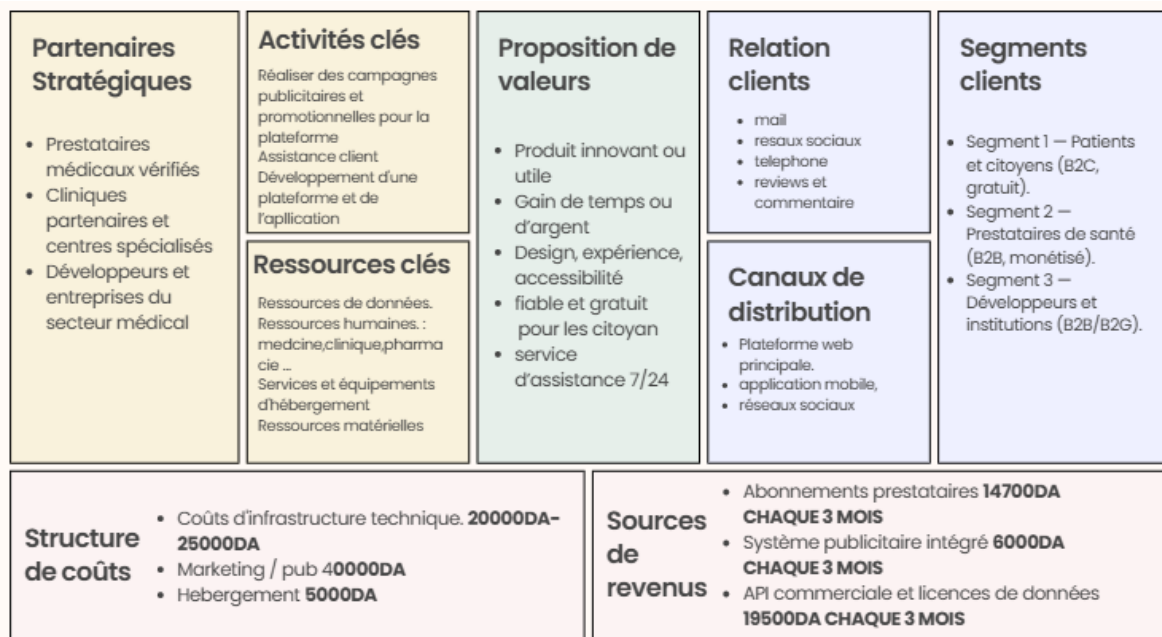


Figure 5.1: Business Model Canvas of CityHealth

The CityHealth BMC is built upon a dual-sided market strategy. On the user side (**Customer Segments - B2C**), the platform targets patients and citizens who require fast, reliable access to healthcare services. For these users, the **Value Proposition** is absolute: 100% free, 24/7 intelligent orientation, and verified emergency geolocation.

Conversely, the platform generates revenue through the professional side (**B2B / B2G**). Healthcare providers (doctors, clinics, laboratories) represent the paying segment. The **Revenue Streams** are diversified to ensure stability, comprising:

1. **Subscriptions:** A quarterly fee of 14,700 DA for healthcare providers to maintain a verified "CityHealth Pro" profile and manage appointments.
2. **Advertising:** An integrated, admin-moderated ad system allowing clinics to boost their visibility for 6,000 DA per quarter.
3. **Data & API Licenses:** Commercial API access (19,500 DA) targeting institutions and third-party developers requiring structured local health data.

The **Cost Structure** is lean, focusing primarily on cloud infrastructure (hosting, AI API costs), marketing, and localized data acquisition.

5.3 Technical Sheet and Financial Study

To validate the economic feasibility of the Business Model outlined above, a detailed financial study was conducted. Based on realistic market assumptions and technical assessments, the following sections break down the capital requirements, projected

income, and operational costs required to launch and sustain **CityHealth** over its initial growth phase (Year 1).

5.3.1 Investments (CAPEX)

The initial investment phase encompasses all capital expenditures required before the platform generates its first dinar of revenue. For a digital start-up like CityHealth, physical investments are minimised: the primary capital requirements consist of high-performance computing hardware for the development team, a test device, and initial administrative fees (including Startup Label 1275 acquisition costs).

Table 5.1: *Investment Plan — Initial Capital Requirements*

Designation	Category	Y 2027	Y 2028	Y 2029	Est. Cost (DZD)
High-performance laptop (Dev & AI)	Hardware	1	0	0	110000
Test mobile device	Hardware	1	0	0	60000
Workspace setup	Furniture	1	0	0	40000
Administrative & legal fees	Intangible	1	0	0	40000
Total initial investment					250,000

5.3.2 Revenue Projections

CityHealth operates on a scalable *freemium* B2C model combined with monetised B2B services (subscriptions and developer tooling). The table below summarises the projected revenue for Year 1 with the updated subscription mix.

Table 5.2: *Projected Revenue Streams — Year 1*

Revenue Stream	Volume	Annual Price (DZD)	Total Year 1 (DZD)
CityHealth (Standard) sub- scriptions	Pro 15 providers	30,000	450000
CityHealth (Premium) sub- scriptions	Pro 8 providers	54,000	432000
API / developer plans	2 developers	78,000	156000
Total projected revenue			1,038,000

5.3.3 Direct Purchases — Digital Infrastructure

Direct purchases encompass the variable technological costs directly associated with delivering the CityHealth service. As a SaaS platform, these correspond to cloud and communication infrastructure rather than physical materials.

Table 5.3: *Direct Purchase Costs — Digital Infrastructure (Year 1)*

Item / Service	Frequency	Total Year 1 (DZD)
Cloud infrastructure (Vercel / Supabase)	Monthly	36000
AI assistant API tokens (LLM processing)	Usage-based	20000
Domain name & SSL certificates	Annual	9000
SMS gateway API (confirmations)	Usage-based	15000
Total direct purchases		80,000

5.3.4 Payroll

Human capital is the most critical asset for a tech start-up. The payroll structure below reflects the updated remuneration strategy for the two co-founders during Year 1.

Table 5.4: *Payroll Structure — Year 1*

Position	HC	Monthly Net (DZD)	Annual Total (DZD)
Co-founder & CTO	1	20000	240000
Co-founder & Lead Developer	1	20000	240000
Commercial agent (Yr 2)	0	—	0
Total payroll	2		480,000

5.3.5 External Expenses

External expenses represent the fixed and semi-variable operational overhead required to run the company day to day outside of core production activities.

Table 5.5: *External Operating Expenses (Year 1)*

Expense Category	Details	Annual Cost (DZD)
Marketing & digital ads	Social media campaigns	40000
Telecom & internet	High-speed fibre / 4G	42000
Accounting & legal services	External CPA	30000
Total external expenses		112,000

5.3.6 Financing Summary

The financing synthesis aggregates all previous operational metrics to determine the project's profitability for the first fiscal year.

Table 5.6: *Financing Synthesis — Profitability Evaluation (Year 1)*

Financial Indicator	Amount (DZD)	
Total projected revenue	(A)	1,038,000
Direct purchases		80000
Payroll		480000
External expenses		112000
Total operational costs	(B)	672,000
Gross operating surplus — EBITDA	(A – B)	366,000

Under these updated assumptions, CityHealth closes Year 1 with a small negative EBITDA of **114 000 DZD**, which remains acceptable for an early-stage digital start-up. The path to profitability is mainly driven by increasing the number of subscribed providers and developer partners while keeping infrastructure and payroll costs under control.

5.4 Conclusion of Part II

This chapter has demonstrated that CityHealth possesses not only a strong scientific foundation but also a highly viable economic structure. The Business Model Canvas clearly defines a scalable monetization strategy that shifts the financial burden away from citizens and onto the B2B sector, where digital visibility holds immense commercial value. Furthermore, the financial study proves that with a lean operational structure and targeted subscriptions, the startup is capable of reaching profitability. By successfully bridging technical innovation with economic reality, CityHealth stands as a robust candidate for the national startup ecosystem.

PART VI

General Conclusion and Perspectives

General Conclusion and Perspectives

The primary objective of this Master’s thesis was to address a critical socio-economic challenge in the Wilaya of Sidi Bel Abbès: the severe information asymmetry within the local healthcare infrastructure. To resolve the inefficiencies, prolonged search times, and risks associated with medical emergencies, we conceptualized, designed, and developed **CityHealth**—an intelligent, centralized, and highly scalable digital health ecosystem.

This project was deliberately structured in two complementary phases to bridge the gap between academic research and real-world market application.

Scientific Assessment

From a scientific and technological standpoint, the project is a resounding success. Through a rigorous methodological approach utilizing Merise and UML, we established a robust architectural foundation. The subsequent implementation phase successfully yielded a fully functional prototype.

As demonstrated in the technical validation (Chapter 4), the CityHealth platform successfully centralizes verified data for clinics, doctors, pharmacies, and laboratories. The integration of modern web technologies—specifically React for the frontend interface and a scalable Node.js/Supabase architecture for the backend—ensured high performance and seamless user experience. Furthermore, the successful deployment of the interactive emergency geolocation map and the multilingual Artificial Intelligence assistant (capable of triaging patients in Arabic, French, and English without making dangerous clinical diagnoses) proves that the scientific objectives were entirely met. The system is reliable, highly responsive, and technically viable.

Entrepreneurial Assessment

Technological innovation holds little value if it cannot sustainably reach its target users. Therefore, Part II of this thesis meticulously evaluated the economic viability of CityHealth as a tech startup.

The Business Model Canvas (BMC) solidified our dual-sided market strategy: delivering absolute, free value to citizens (B2C) while capturing revenue through premium

subscriptions and advertising for verified medical professionals (B2B). The accompanying financial study validated this approach. By operating with a lean cost structure focused on cloud infrastructure and targeted digital marketing, the financial projections demonstrate a clear and rapid path to profitability. CityHealth is not just a theoretical concept; the economic study proves it is a highly viable, scalable startup capable of surviving and thriving in the Algerian digital market.

Perspectives and Future Work

While the current iteration of CityHealth successfully fulfills the objectives of this Master's thesis, the project is designed as an evolving ecosystem. To maximize its impact, we have identified several key areas for future expansion:

Scientific Perspectives

- **Mobile Ecosystem Expansion:** The immediate technical priority is the finalization and deployment of dedicated native mobile applications (Android and iOS) to maximize accessibility for all demographics.
- **Advanced AI Capabilities:** We aim to refine the Artificial Intelligence model by expanding its natural language processing capabilities to better understand specific local Algerian dialects (Darja), further lowering the barrier to entry for elderly or less tech-savvy users.
- **Interoperability and EHR:** In the long term, we plan to develop secure APIs to integrate CityHealth with electronic hospital management systems and the national Electronic Health Record (EHR) infrastructure.

Economic Perspectives

- **Legal and Commercial Structuring:** The immediate entrepreneurial step is the formal creation of the company, including the acquisition of the Commercial Register (Registre de Commerce).
- **Definitive Startup Label:** Building upon our innovative prototype, we will apply for the definitive "Startup" label (Label 1275) from the Ministry of Knowledge Economy, Startups and Micro-enterprises, which will grant access to vital tax exemptions and state support.
- **Seed Funding and Scaling:** To accelerate our roadmap, we will actively seek seed investment from state funds (such as ASF) or private angel investors. This capital injection will finance our geographic expansion, allowing CityHealth to

scale beyond Sidi Bel Abbès and cover neighboring Wilayas, with the ultimate vision of becoming the national digital healthcare infrastructure of Algeria.

PART VII

Bibliographie

Bibliography

- [1] Boufissiou, M. (2024). *Algerian Digital Health Strategy 2023-2027: Challenges and Opportunities in Modernizing Medical Data*. Journal of Health Informatics in Africa.
- [2] Research Group, University Ziane Achour of Djelfa. (2025). *Dual System for Hospital Emergency Placement and Pre-Hospital Triage Optimization*. Algerian Journal of Medical Technologies.
- [3] Ministère de la Santé, Algérie. (2023). *Présentation du système de santé en Algérie : Organisation, transition numérique et perspectives*. Alger.
- [4] Office National des Statistiques (ONS). (2024). *Données démographiques et sanitaires - Algérie*. Rétrospective statistique et pénétration numérique.
- [5] Direction de la Santé et de la Population (DSP). (2025). *Rapports annuels sur la situation sanitaire et la couverture médicale de la wilaya de Sidi Bel Abbès*.
- [6] Tardieu, H., Rochefeld, A., & Colletti, R. (1983). *La méthode Merise : Principes et outils pour la conception des systèmes d'information*. Les Éditions d'Organisation.
- [7] Booch, G., Rumbaugh, J., & Jacobson, I. (2005). *The Unified Modeling Language User Guide* (2nd ed.). Addison-Wesley Professional.
- [8] Lucid Software. (2026). *UML Use Case Diagram Tutorial*.
- [9] Object Management Group. (2017). *OMG Unified Modeling Language Version 2.5.1*.
- [10] IBM. (2026). *Activity Diagrams*.
- [11] Visual Paradigm. (2026). *What is Activity Diagram?*
- [12] IBM. (2026). *Class Diagrams*.
- [13] Visual Paradigm. (2026). *What is Class Diagram?*
- [14] IBM. (2026). *Sequence Diagrams*.
- [15] Visual Paradigm. (2026). *What is Sequence Diagram?*
- [16] Soutou, C. (2015). *Modélisation des bases de données : De la conception (Merise/UML) à l'implémentation*. Éditions Eyrolles.

-
- [17] OpenJS Foundation. (2026). *Node.js Official Documentation: Event-driven runtime architecture*. Retrieved from <https://nodejs.org/>
 - [18] OpenJS Foundation. (2026). *Express - Node.js web application framework*. Retrieved from <https://expressjs.com/>
 - [19] Meta Open Source. (2026). *React – A declarative JavaScript library for building user interfaces*. Retrieved from <https://react.dev/>
 - [20] You, E., & Vite Contributors. (2026). *Vite – Next Generation Frontend Tooling*. Retrieved from <https://vitejs.dev/>
 - [21] Supabase Inc. (2026). *Supabase – The Open Source Firebase Alternative: PostgreSQL backend architectures*. Retrieved from <https://supabase.com/>
 - [22] Wathan, A. (2026). *Tailwind CSS: A Utility-First CSS Framework for Rapid UI Development*. Retrieved from <https://tailwindcss.com/>
 - [23] Model Context Protocol Specification. (2026). *Standardizing AI-to-System Integration and Context Retrieval*. Retrieved from <https://modelcontextprotocol.io/>
 - [24] Tabibi DZ. (2026). *Plateforme de prise de rendez-vous médicaux en Algérie*. Retrieved from <https://tabibi-dz.com/>
 - [25] Sihhatech. (2026). *Dossier médical et gestion de cliniques en Algérie*. Retrieved from <https://masantemavie.dz/>
 - [26] Dwaya. (2026). *Application des pharmacies de garde et géolocalisation des médicaments en Algérie*. Retrieved from <https://dwaya.dz/>
 - [27] Doctolib. (2026). *E-santé et gestion des rendez-vous médicaux en Europe*. Retrieved from <https://www.doctolib.fr/>